CHEMICAL ENGINEERING DEPARTMENT: BACHELOR OF CHEMICAL ENGINEERING

REVISED SYLLABUS OF BACHELOR OF CHEMICAL ENGINEERING 2024

2ND YEAR 1ST SEMESTER

Basic Science	**Course Code: FET/BS/B/Math/T/211**	**MATHEMATICS - III**	**Credits: 3**
	Total Contact Hours: 36		
Prerequisite			
	Course Content		**Hours**
Module V **Module VI**	Fourier Series and Integral Transforms: Fourier series; Periodic functions; Trigonometric series of sine and cosines; Euler's formula; Even and odd functions; Dirichlet's conditions; Half range sine and cosine series; Fourier transform, definitions and properties; Inverse Fourier transform; Convolution; Laplace transform; Convolution; Z transform and properties. Ordinary Differential Equation (ODE) and Series solutions: First order exact differential equation and first order linear differential equation; Second and higher order linear differential equations with constant coefficients; Euler and Cauchy equation; Method of variation of parameters; Ordinary point and regular singularity of a second order linear differential equation; Series solutions; Solution of Legendre and Bessel's equations; Generating functions; Recurrence relations and their Orthogonal properties. Partial Differential Equation (PDE): First order PDE; Lagrange method; Second order PDE with constant coefficients and their classification to Elliptic, Parabolic and Hyperbolic type, Solution of PDE by method of separation of variables; Solution of one-dimensional wave and diffusion equation; Laplace equation of two dimensions.		
Course Outcomes (COs):	The students of the course should be able to – CO1: Define and visualize the concept of probability and statistics CO2: Understand and solve the problems related to vector analysis and differential equations CO3: Learn and apply different techniques of solving ordinary and partial differential equations. CO4: Apply and analyse the concept of different theorems in solving various problems.		
CO-PO Mapping			

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	**1**	**3**										
CO2	**1**	**3**	**2**	**1**								
CO3	**1**	**3**	**2**									
CO4	**1**	**3**	**2**	**2**								

Engineering Science	Course Code: Che/ES/B/Mech/T/212	ENGINEERING THERMODYNAMICS	Credits: 3
	Total Contact Hours: 36		
Prerequisite			
	Course Content		**Hours**
	Basic concepts: Systems-isolated, closed and open; thermodynamic state; state variables-intensive and extensive; process-cyclic process, reversible process; Zeroth law of thermodynamics-concept of temperature, properties of pure substances, energy interaction between system and surroundings-work and heat; adiabatic and isothermal processes. Concept of internal energy; conservation of energy; first law of entropy; entropy change of the universe-possible and impossible processes; reversible and irreversible processes; internal and external reversibility; thermodynamic temperature. Open systems- first and second law applications; second law efficiency and availability, irreversibility, lost work and energy. Third Law of thermodynamics. Thermodynamic properties- enthalpy, Gibb's free energy and Helmholtz free energy; application of first and second law to closed systems; heat engines - Carnot cycle; Introduction to refrigeration and gas liquefaction processes, introduction to power cycles.		
Course Outcomes (COs):	The students of the course should be able to – CO1: Define thermodynamic system and related terminologies (K1) CO2: Interpret the fundamental laws and principles (K2) CO3: Apply laws and principles of thermodynamics for simple engineering devices (K3) CO4: Analyze power generation and refrigeration systems using thermodynamic principles (K4)		
CO-PO Mapping			

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1										
CO2	3	2	1									
CO3	2	3	1									
CO4	1	3	2	1	1							

Program CORE	Course Code: Che/PC/B/T/213	MECHANICS OF FLUID	Credits: 3
	Total Contact Hours: 36		
Prerequisite			
	Course Content		**Hours**
	Introduction: Fluid, Concept of continuum, Method of analysis: System and Control volume, Methods of description - Lagrangian and Eulerian Fundamental Concepts: Velocity field, Streamlines, Streaklines, Pathlines, Timelines, Stress field, Viscosity, Newtonian fluid and Non-newtonian fluids, Description and Classification of fluid flow-inviscid viscous, laminar, turbulent, internal, external, incompressible, compressible Fluid Statics: Pressure variation in a static fluid Control volume analysis (integral approach): - Reynolds transport theorem, conservation of mass and momentum theorem, The first and second law of Thermodynamics Differential analysis of fluid motion: Conservation of mass; Stream function for two-dimensional incompressible flow, Kinematics of flow field: Acceleration of a particle in a Velocity Field, Fluid Rotation, Vorticity, Fluid Deformation. Momentum equation: Forces acting on a Fluid particle, Differential momentum Equation Incompressible Inviscid flow: Momentum Equation for frictionless flow, Euler's equation, Bernoulli equation, Static, Stagnation and Dynamic pressures. Incompressible Viscous flow, pipes and channels - laminar and turbulent flow, nature of turbulence, deviating velocities in turbulent flow, Intensity and scale of turbulence, Reynolds stress, Prandtl mixing length, boundary layer, Universal velocity distribution, corrections to Bernoulli's equation, Friction factor, Friction factor vs Reynold number, flow through circular and noncircular conduits, Flow in agitated vessels, Hagen Poiseuille equation, Fanning equation, major and minor frictional losses, skin friction and form friction, pressure drop in flow through multiple paths, pipeline design. Case studies on pipeline design, selection of appropriate pump characteristics for a specific pipeline External flows: Boundary layer Drag and lift, drag coefficient, Stokes law, Drag coefficient, Drag coefficient vs Reynold s number plot Flow measuring devices: design considerations Case studies on design of venturimeter, orificemeter, rotameter Flow through packed bed and fluidized bed: case study on flow of water through ion exchange packed bed.		
List of Text Books/ Reference Books/Open sources	1. Introduction to Fluid Mechanics – Fox and McDonald 2. Fundamentals of Fluid Mechanics – Philip M. Gerhart, Richard J. gross, John I. Hochstein 3. Fluid Mechanics – Victor L. Streeter 4. Fluid Mechanics and its Application – Vijay Gupta & Santosh K. Gupta		
Course Outcomes (COs):	On successful completion of this course students will be able to CO1: Identify types of fluids and demonstrate their fundamental concepts of continuum fluid mechanics and flow visualization techniques. CO2: Solve problems applying mass, momentum, mechanical energy and total energy balance equations for integral and differential control volumes. CO3: Analyze and solve a range of Chemical Engineering operational and design problems for flow through pipelines, agitated vessels, packed and fluidized beds. CO4: Design of manometers and fluid flow measuring devices applying fundamental knowledge of fluid statics and fluid dynamics.		

CO-PO Mapping

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3
CO1	3												3		
CO2	3	3											3		
CO3	3	3	3	3									3	3	
CO4	3	3	3										3	3	

Engineering Science	Course Code: Che/ES/B/Mech/T/215	STRENGTH OF MATERIALS	Credits: 3
	Total Contact Hours: 36		
Prerequisite			
	Course Content		**Hours**
	Uniaxial stress field, Thin pressure vessels, Torsion (inclusive of Helical spring), shear force and Bending moment, Bending and shear stress in beams, Deflection beams, Energy methods in Strength of Materials, Problem of Plane stress and strain, Theories of failure, Buckling of columns.		
Course Outcomes (COs):	Students will be able to CO1: Solve simple engineering problems under uniaxial, torsional and bending loading. (K2) CO2: Apply the concepts to approach general engineering problems for biaxial stress field. (K2) CO3: Apply the concepts to approach engineering problems related to buckling of columns. (K2) CO4: Assess the fundamental mechanical behaviour related to mechanics of deformable bodies. (K3)		
CO-PO Mapping			

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	1	1		1				1		
CO2	3	3	2	2		1				1		
CO3	3	3	1	1		1				1		
CO4	2	2	1	1		1				1		

Basic Science	Course Code: Che/BS/B/Chem/T/214	PHYSICAL CHEMISTRY	Credits: 3
	Total Contact Hours: 36		
Prerequisite			
	Course Content		**Hours**
	Electrochemistry: Electrolytic conductance, Transference Number, Specific and Equivalent Conductance (Debye-Huckel, Onsager relations (no derivations), absolute velocity of ions, degree of ionization and conductance, application of conductance (solubility of sparingly soluble salts, conductometric titrations), concept of activity and activity coefficient Cells (reversible and irreversible cells), electromotive force, standard cells, cell relation and electromotive force (EMF), convention on the sign of EMF, single electrode potential, calculation of cell EMF from single electrode potential, thermodynamics of electrode potentials (electromotive series, standard electrode potentials) equations for cell EMF, standard potentials and equilibrium constants, types of electrodes (hydrogen and calomel electrodes in particular) Electrochemical Cells Application: solubility product and EMF, potentiometric determination of pH, potentiometric titration, electrolysis and polarization, decomposition potential, overvoltage (its measurement), electrolytic separation of metals, commercial cells, fuel cells. Surface Chemistry: Surface tension, surface energy, flat and curve surface, pressure differential across a curved surface and capillary rise and fall, Kelvin equation, influence of soap molecules on surface tension, Gibbs monolayer and Gibbs equation (derivation optional) and surface energy micro heterogeneous systems and their uses (micelles, emulsions, microemulsions with reference to the basis of their formation, vesicles and lipozomes), contact angle, wetting, spreading and adhesion, adsorption and colloidal dispersions, types of colloids, colloid solubility, concept of zeta potential and electrokinetics (electro osmosis, electrophoresis, straining potential, sedimentation potential), nano dispersions, their preparation, synthesis and uses.		
Course Outcomes (COs):	Students will be able to **CO1**: Define and explain conductance and related physical quantities and application of conductance measurements **CO2**: Explain the interconversion of chemical and electrical energy and develop the concept and construction of various types of cells and their cell potentials **CO3**: Apply EMF measurement methods to determine various electrochemical parameters **CO4**: Explain the origin of surface tension and its various manifestations along with the basics of micelles, emulsions, microemulsions and their applications **CO5**: Discuss the details of colloids, nanomaterials and their applications in various areas and the various types of electrokinetic phenomena		
CO-PO Mapping			

Program Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO9	PO10	PO11	PO12
CO1	3	3											
CO2	3	3											
CO3	3	3											
CO4	3	3											
CO5	3	3											

Program CORE	Course Code: Che/PC/B/T/216	CHEMICAL PROCESS PRINCIPLES	Credits: 3
	Total Contact Hours: 36		
Prerequisite			
	Course Content		**Hours**
	Features of organized chemical processing- from Chemistry to Chemical Engineering, Chemical Industry- scope, features & characteristics. Unit Operations in Chemical Engineering - an overview. Units and Dimensions - Systems of units, Conversion of units, Dimensional homogeneity and consistency, dimensional analysis, dimensionless groups and their use in chemical engineering. Concepts of chemical similitude and scale up. Material balances: Fundamentals; Batch and flow processes, Steady-flow and unsteady processes, material balance calculations for single-unit and multiple-unit processes, material balance for reactive and non-reactive systems, processes with recycle and/or bypass with or without purge, material balance for combustion reactions, unsteady-state and transient processes. Group discussion on the development of flowchart using a case study of hydrogenation of propane in a reactor with recycle; using preheating of feed using heat exchanger, cooling of product stream using a cooler; recovery of products and unreacted reactants using a sequence of absorber, stripper and a distillation tower Material balances: Process Classification, Continuous steady-state balance, Integral balance on batch and semi-batch process, Flowchart, Degrees of freedom analysis, Single unit and multi-unit material balance equation, Recycle and Bypass. A case study on Air Conditioner. Group discussion on the mass balance of non-reactive system using the case study on distillation tower used for the separation of benzene and toluene. Material balance on reactive systems – Limiting and excess reactant, fractional conversion, chemical equilibrium, extent of reaction, yield and selectivity, overall and single pass conversion, Balances on molecular and atomic species, reactive system , reactive system with recycle and purge, Combustion reactions - wet and dry basis, theoretical and excess air. . A case study on the production of methanol from hydrogen and carbon dioxide. Energy balances on nonreactive processes: Introduction to energy balances, energy balances on closed and open systems - the steady-flow energy equation, mechanical energy balances, Energy balances for phase change operations, heat of mixing and heat of solution - energy balance for mixing and dissolution processes. Case study on steam turbine. Energy balances on reactive processes: energy balance in reactors under steady and unsteady state conditions; combustion of fuel (concepts of higher and lower heating values of fuel) Combined material and energy balances including concepts of Psychrometry (definitions of absolute and relative humidity, dry and wet bulb temperatures, humid heat, humid volume), problems on humidification, dehumidification. Some typical industrial problems requiring simultaneous mass and energy balance, e.g., problems on dryer, cooling tower etc and problems with reactive changes. Case study on humidification of air in an adiabatic spray tower. Unsteady state material and energy balances with special reference to batch and semi-batch reactive and non-reactive processes. Group discussion on the mass and energy Balances on reactive system using the case study on dehydrogenation of ethanol to acetaldehyde.		
List of Text Books/ Reference Books/Open sources	1. Elementary Principles of Chemical Processes, R. M. Felder and R. W. Rousseau. 2. Chemical Process Principles, O. A. Hougen, K. M. Watson, R. A. Ragatz. 3. Basic Principles and Calculation in Chemical Engineering by David M. Himmelblau.		
Course Outcomes (COs):	On successful completion of this course students will be able to CO1 Define, recognize and examine Unit Operations and processes, dimensional homogeneity and consistency, batch and continuous systems and steady and unsteady states (K1,A1,A2) CO2 Apply dimensional analysis methods to develop and relate dimensionless groups, analyze chemical similitude and use dimensionless groups in scale up (K3, K4, K6, A4) CO3 Prepare flowcharts to represent Chemical processes and apply degrees of freedom analysis (K3,A5)		

	CO4 Analyse flowcharts and develop individual and combined mass and energy balance equations to evaluate unknown values of mass and energy flows for steady and unsteady state non-reactive systems (K4, K6, A3) CO5 Analyse flowcharts and develop individual and combined mass and energy balance equations to evaluate unknown values of mass and energy flows for steady and unsteady state reactive systems (K4, K6, A3)
CO-PO Mapping	

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PS O 1	PS O 2	PS O 3
CO1	3	3							3			1	3	3	3
CO2	3	3	3	3					3			2	3	3	3
CO3	3	3	3	3					3			2	3	3	3
CO4	3	3	3	3					3			2	3	3	3
CO5	3	3	3	3					3			2	3	3	3

Sessional 1	Course Code: Che/BS/B/Chem/S/211	PHYSICAL CHEMISTRY LABORATORY	Credits: 2
	Total Contact Hours:		
Prerequisite			
	Course Content		**Hours**
Experiment 1	Determination of viscosity coefficient		
Experiment 2	Determination of surface tension		
Experiment 3	Determination of distribution coefficient		
Experiment 4	Determination of equilibrium constant (homogeneous)		
Experiment 5	Determination of phase diagram (ternary system)		
Experiment 6	Determination of adsorption isotherm.		
Experiment 7	Conductometric titration		
Experiment 8	Potentiometric / pH-metric titration		
Experiment 9	Verification of Beer's law		
Experiment 10	Determination of specific rotation		
Experiment 11	Determination of CMC		
Experiment 12	Analytical application of refractive index measurement		
Experiment 13	Determination of decomposition voltage		
Course Outcomes (COs):	CO1: Develop basic concepts of preparing primary and secondary standard solutions in the laboratory and apply theoretical knowledge to determine the strength of solutions of unknown concentration CO2: Determine physical constants using simple glass apparatus and instruments Explain how one can determine the strength of solutions of unknown concentration by conductometric titration		
CO-PO Mapping			

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO9	PO10	PO11	PO12
CO1	3	3											
CO2	3	3											

Sessional 2	Course Code: Che/ES/B/Mech/S/212	WORKSHOP PRACTICE – XII	Credits: 2
	Total Contact Hours:		
Prerequisite			
	Course Content		**Hours**
	Introduction to machine tools - lathes, drilling machines, shaping machines, planning machines, slotting machines, milling machines, grinding machines; machine shop work involving different operations by using the above mentioned machines through making of jobs. Experiments on: Study of the speed structure of a lathe, study of apron mechanism and calibration of feeds in a lathe. Study and grinding of various cutting tools.		
Course Outcomes (COs):	The students of the course should be able to CO1: Discuss different types of machine tools like lathes, drilling machines, shaping machines, planning machines, slotting machines, milling machines, grinding machines. (K2) CO2: Perform various machining operations on Lathe to manufacture a pin (S2) CO3: Perform various machining operations on shaping machine to manufacture a parallel block. (S2) CO4: Perform various machining operations on milling machine to manufacture a spur gear (S2) CO5: Calibrate the performance of apron mechanism, speed structure and feed of a lathe. (S3)		
CO-PO Mapping			

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1								2	1	
CO2	2	1				2			2	1		
CO3	2	1				2			2	1		
CO4	2	1				2			2	1		
CO5	3	2				2			2	1		

Sessional 3	**Course Code: Che/ES/B/Mech/S/213**	**COMPUTER AIDED DRAFTING**	**Credits: 1.5**
	Total Contact Hours:		
Prerequisite			
	Course Content		**Hours**
	Introduction to a computer aided drafting software, Basic commands of 2D drafting, Drafting assignment. Concept of Layer, Layout, Model space, Paper space, Viewport, Creation and use of template, Drafting assignment. Dimensioning, Blocks, Attributes, accessing internal and external database files, Drafting assignments. Isometric drawing using iso-planes, Drafting assignment.		**9** **12** **12** **9**
Course Outcomes (COs):	The students of the course should be able to – CO1: Identify the basic 2D drafting tools (drawing and editing) and toolbars in drafting software. (K1, A1, S1) CO2: Interpret the concept of layer, and of model space and paper space in conjunction with viewport. (K2, A2, S2) CO3: Describe the concept of blocks, attributes etc. to better organize complicated drawings. (K2, A2, S2) CO4: Prepare engineering drawing of simple machine components up to its paper printout with proper scale. (K3, A3, S3) CO5: Construct isometric drawings for simple engineering components. (K3, A2, S2)		
CO-PO Mapping			

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1			1		3	1		1	2	1		
CO2			1		3	1		1	2	1		
CO3			1		3			1	1	1		
CO4	2		2		3	1		1	2	2		
CO5	2		1		3	1		1	2	2		

CHEMICAL ENGINEERING DEPARTMENT: BACHELOR OF CHEMICAL ENGINEERING

REVISED SYLLABUS OF BACHELOR OF CHEMICAL ENGINEERING 2024

2ND YEAR 2ND SEMESTER

Program CORE	**Course Code: Che/PC/B/T/221**	**Numerical Analysis for Chemical Engineers**	**Credits: 3**
	Total Contact Hours: 36		
Prerequisite			
	Course Content		**Hours**
	Linear Algebraic Equations: Solution of simultaneous linear algebraic equations, Gaussian elimination, Thomas algorithm, Determination of inverse of a matrix: Gauss-Jordan method;		5
	Iterative solution of a set of simultaneous linear algebraic equations: Jacobi method, Gauss Seidel method, Relaxation method.		3
	Nonlinear algebraic equations: Single nonlinear equation: Bisection method, Successive substitution, Secant (Regula falsii) method, Newton-Raphson method, Determination of roots of a polynomial, Solution of simultaneous nonlinear algebraic equations: Successive substitution, Newton-Raphson method.		5
	Regression: Method of Least Squares, Linear and Nonlinear Least Squares. (2pds)		2
	Interpolation: Newton's Divided Difference Formulae, Lagrange Interpolation; Equispaced base-point methods - Newton's Forward Difference and Backward Difference Formula.		4
	Numerical Differentiation - based on interpolation formulae. (1 pds)		1
	Numerical Integration - Newton-Cotes formulae, Trapeziodal Rule, Simpson's Rule, Composite formulae		2
	Solution of Ordinary Differential Equations - Initial Value Problem (IVP): Explicit methods: Adam-Bashforth techniques; Implicit methods: Adams-Moulton techniques; Predictor-Corrector Formulae; Runge-Kutta and Runge-Kutta-Gill methods.		4
	Solution of Ordinary Differential Equations - Boundary Value Problem (BVP): Shooting Method, Finite Difference Methods		3
	Solution of Partial Differential Equations: Elliptic, Hyperbolic and Parabolic PDEs; Nonlinear PDEs, Laplace's equation		4
Course Outcomes (COs):	The students of the course should be able to: CO1: Understand numerical methods to solve problems involving various chemical engineering operations (K2) CO2: Determine solution of simultaneous linear equations and root of non-linear equations (K5) CO3: Solve problems using regression analysis, interpolation, and differentiation and integration numerically. (K3) CO4: Solve chemical engineering related problems using numerical solution of ordinary differential equations. (K4) CO5: Solve chemical engineering related problems using numerical solution of partial differential equations. (K4)		
CO-PO Mapping			

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3														
CO2	3	2	3	3	2							1		2	
CO3	3	2	3	3	2							1		2	
CO4	3	3	3	3	2							1			3
CO5	3	3	3	3	2							1			3

Engineering Science	Course Code: Che/ES/B/Met/T/222	MATERIAL SCIENCE AND ENGINEERING	Credits: 3
	Total Contact Hours: 36		
Prerequisite			
	Course Content		**Hours**
	Metal extraction processes: Unit processes involved in extraction of metals from oxide and sulphide ores; Flow sheets for iron & steel, zinc, copper, lead, aluminum, titanium and nickel Materials for construction pertaining to chemical processing and equipment: Classification –metals & alloys, ceramics, polymers, composites, nanostructured materials. Properties – corrosion resistance, thermal resistance, catalytic properties Specific application in Chemical Industries		
List of Text Books/ Reference Books/Open sources	• Extraction of Nonferrous metals – H. S. Ray, R. Sridhar, K.P Abraham • Materials Science and Engineering: An Introduction – William D. Callister, Jr, John Wiley & Sons, Inc. Materials Science and Engineering: A First Course – V. Raghavan, PHI Learning Pvt. Ltd.		
Course Outcomes (COs):	CO1: Understanding Mineral Beneficiation and Extraction of Metals from Oxide and Sulphide Ores Fundamentals (K1,K2). CO2: To know the importance and processes involved in extraction of metals (K3). CO3: To know the different types of metals, alloy and materials for construction pertaining to chemical processing and equipment (K1). CO4: Understand different Properties of materials and its application in chemical industries (K2,K4).		
CO-PO Mapping			

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	2	2	3	3	3	3	3	2	1	3			
CO2	3	3	2	3	3	1	3	2	2	2	1	3			
CO3	3	3	3	3	3	2	1	2	2	2	1	3			
CO4	3	3	3	3	3	1	2	2	3	2	2	3			

Program CORE	Course Code: Che/PC/B/T/223	CHEMICAL ENGINEERING THERMODYNAMICS	Credits: 3
	Total Contact Hours: 36		
Prerequisite			
	Course Content		**Hours**
	Application of thermodynamics to flow processes: flow through ducts, pipes, nozzle for compressible fluids; turbines, compression processes; ejectors. Project work on the flowing influences through devices for compressible fluid for student activity. Volumetric Properties of pure fluids: Cubic Equations of State (EOS); Virial EOS, Benedict-Web-Rubin EOS; Generalized compressibility factor correlation – acentric factor; Lee-Kesler method; Generalized virial coefficient correlation. Thermodynamic property relations for homogenous phase of constant composition: Maxwell Relations, Relation between heat capacities; Relation of thermodynamic potentials with measurable properties; Project work on the evaluation of Thermodynamic properties for student activity. For pure species - Clapeyron equation; Kirchhoff relation, Residual properties and estimation from EOS, thermodynamic diagrams; fugacity and fugacity coefficient and estimation from EOS, Vapor/Liquid equilibrium, fugacity of compressed liquid; extension to gas mixtures. Solution Thermodynamics: partial properties, relation between molar and partial properties, Gibbs'-Duhem equation, relation among partial properties, partial properties in binary solution; chemical potential, Ideal gas mixtures; fugacity coefficient of species in solution and estimation, Ideal solution model, Lewis-Randall rule, Henry's Law; Excess property relations, Excess property and activity coefficient, nature of excess properties, Excess Gibbs free energy models; property changes of mixing; heat effects of mixing process. Phase Equilibria: equilibrium and stability, chemical potential and phase equilibria; Gibbs' phase rule; Vapor/Liquid equilibrium (VLE)- basic equations for low and high pressures VLE, Raoult's Law, Bubble point and dew point calculations; binary systems- Equilibrium for miscible, partially miscible and immiscible systems; estimation of excess Gibbs free energy model parameters from experimental data; Liquid/Liquid equilibrium; Solid/liquid equilibria; Equilibrium adsorption of gases on solid; Osmotic pressure. Chemical Reaction Equilibria: Extent of reaction; Equilibrium constant; Effect of operating conditions on equilibrium conversion; determination of equilibrium compositions for homogeneous gas-phase reactions - single and multiple reactions; liquid-phase reaction equilibria; phase rule for reacting systems.		
List of Text Books/ Reference Books/Open sources	1. Introduction to Chemical Engineering Thermodynamics by J.M. Smith, H C Van Ness, M M Abbott. **2.** Chemical Engineering Thermodynamics by Y V C Rao		
Course Outcomes (COs):	Students will be able to CO1: Students will be able to estimate properties of substances CO2: Students will be able to solve problems on physical and chemical equilibrium CO3: Use available thermodynamic data for applications in chemical engineering processes		
CO-PO Mapping			

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PS O 1	PS O 2	PS O 3
CO1	3												3		
CO2	3	3											3		
CO3	3	3	3	3									3	3	
CO4	3	3	3										3	3	

Program CORE	Course Code: Che/PC/B/T/224	Introduction to Transport Phenomena	Credits: 4
	Total Contact Hours: 48		
Prerequisite			
	Course Content		**Hours**
	Introduction and overview: Importance of heat, mass and momentum transfer in chemical engineering operations using specific examples/Case Studies. Explanation of difference between a thermodynamic process and transport process with a Group Discussion. Overview of transport mechanisms – Diffusion and Convection. Diffusive transport; Convective transport: Natural convection and forced convection. Difference between unit operations and transport processes. Dimensional analysis: usefulness and limitations: Three major classes of non-dimensional numbers based on their significance. Momentum Transport Newtonian and Non-Newtonian fluids: constitutive relationships. Shell momentum balance and its application to determine shear stress and velocity distribution for one dimensional flow. Derivation of equation of continuity, equation of motion, Navier Stokes equation, Euler's equation for Cartesian coordinate based system. Generalized vector form of equation of continuity and Navier Stokes equation. Fluid-solid, liquid-gas/vapor and liquid-liquid boundary conditions for fluid flow systems along with Group Discussions with practical examples. Nondimensionalization of the Navier-Stokes equation; Non-dimensional numbers (Reynolds number, Froude number, Strouhal number , Euler number); Concept of scale up. One dimensional flow under steady-state: Case Studies on i) Laminar incompressible flow of film along an inclined plane, ii) Incompressible laminar flow of fluid through slit made up of two parallel vertical plates, iii) Couette flow, iv) flow through circular pipe, v) flow through annulus. Exact analytical solutions for shear stress profile and velocity profile, determination of average velocity and flow rates, net force acting on a surface for Newtonian fluid, Bingham plastic fluid, Power law fluid. Boundary Layer theory over a flat plate: Energy/ Heat Transport: Derivation of Generalized equation for heat diffusion/conduction in solid. Fourier's law of heat conduction. Thermal Properties. Boundary and initial conditions: One dimensional heat conduction under steady-state: Resistance to heat transport: multiple resistances in series and in parallel: Case studies on analysis for plane wall, cylinders and spheres. Transpiration cooling: Case studies on design of insulation system in pipes and in spherical storage vessels Analysis of steady state heat conduction with heat source: Case studies with electrical, nuclear (nuclear reactor), viscous and chemical heat sources. Case studies on cooling of microprocessor employing fin Transient heat conduction in solid: Lumped capacitance model, Definition of Biot number and validity of lumped capacitance model. Transient heat conduction with spatial Effects: Semi-infinite Solid: Solution using similarity variable/ Laplace transform: expression for temperature distribution and heat flux. Convective transport in flow system: Thermal boundary layer, definition of heat transfer coefficient, Nusselt number. Derivation of generalized two dimensional thermal boundary layer equation (with viscous heat generation) and boundary conditions: Case study on analysis of Couette flow with heat transfer : Exact solution of velocity and temperature profiles.		

Mass Transport:
Definition of diffusion velocity, diffusive flux (j_A) and mass flux with respect to stationary coordinates (n_A), derivation of relationship between n_A and j_A; Fick's first law and Mass Diffusion coefficient. Derivation of generalized species mass diffusion equation, boundary and initial conditions. Steady state Mass Diffusion without chemical reaction: Derivation of concentration profile, average concentration and mass flux: Counter diffusion and Diffusion through a 'stagnant' media. Steady State Mass diffusion with heterogeneous catalytic surface reaction, Steady State Mass diffusion with homogeneous chemical reactions.

Transient Diffusion: Biot number for mass transfer, Lumped capacitance mode, Transient Diffusion with special variation: Fick's 2nd law, Semi-infinite media: Analytical solution using similarity variable/Laplace transform. Convective mass transfer: concentration boundary layer and definition of mass transfer coefficient. Generalized two dimensional concentration boundary layer equations and boundary conditions.

Case studies on oxygen transport from the interior lung cavity across the lung tissue to the network of blood vessels of the human body; transport of drug, dissolution of salt slab under unsteady-state condition; transport of nutrient in intestine consisting of villi.

Boundary Layer Similarity: Velocity, thermal and concentration boundary layers: Approximation and Normalized convection transfer equations : Boundary layer similarity parameters, Functional form of the solution, Physical significance of Dimensionless Numbers, Boundary Layer Analogy: The Momentum, Heat and Mass transport analogy: Evaporative cooling; The Reynolds Analogy, Chilton Colburn Analogy, Prandtl Analogy.

List of Text Books/ Reference Books/Open sources

1. Transport Phenomena by R. Byron Bird, Warren E. Stewart, Edwin N. Lightfoot
2. Momentum Heat and Mass Transfer by C. O. Bennett and J. E. Myres.
3. Fundamentals of Heat and Mass Transfer by Frank P. Incropera, David P. DeWitt

Course Outcomes (COs):

The students of the course should be able to
CO1. Identify the transport properties and describe different mechanisms of momentum, energy and mass transport. (K1, K2, K4,)
CO2. Develop the governing conservation equations and boundary conditions for the steady state and transient momentum, heat and mass transport (K3, K5)
CO3. Analytically solve and analyze a variety of steady state and transient momentum, heat and mass transport problems with appropriate assumptions and approximation (K4,K5)
CO4. Analyze and solve a practical real life problem applying the momentum, heat and mass transport equations and appropriate solution techniques (K4, K5, K6, A2)
CO5. Non-dimensionalize the transport equations, identify the dimensionless numbers and apply analogies between momentum, heat and mass transport to scale up. (K2, K3,K5)

CO-PO Mapping

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3
CO1	3	2											3		
CO2	3	2	1	1									3		
CO3	3	3	1	1									3		
CO4	2	3	3	1	1								3	2	
CO5	2	3	2										2		

Program CORE	Course Code: Che/PC/B/T/225	MECHANICAL OPERATIONS	Credits: 3
	Total Contact Hours: 36		
Prerequisite			
	Course Content		**Hours**
	Particulate Solids, particle characterization, sphericity, particle size distribution, mean particle size of solids. Size Reduction of solids: Crushing, Grinding, Mechanism of size reduction, Energy for size reduction, Kick's law, Rittinger's law, Bond's law, Energy utilization; Method of operating crushers: free crushing, choke feeding, open circuit grinding and closed circuit grinding; Classification of size reduction equipment based on feed size and product size: Coarse crusher: jaw crusher, Blake Crusher, Intermediate crusher: Rolls, angle of nip, determination of roll diameter, fine grinding: Ball mills: factors influencing the size of the product, critical speed of rotation. Size separation & gas cleaning: Classification of different methods of separation based on particle size. Screen analysis Motion of particles in a fluid: Drag, Stokes law regime, Newton's law regime, terminal velocity, free settling, hindered settling, accelerating motion of a particle in the gravitational field and in centrifugal field. Separation of materials by size and by difference in density: Gravity settling, Working principle & design of Elutriator, double cone classifier, spitzkasten, Rake classifiers, Hydraulic Jig; Sedimentation: design of continuous thickener, role of flocculation and agglomeration Centrifugal separators: Working principle and design of Cyclone separator: cyclone efficiency, hydrocyclone, centrifuge Froth flotation and its application Electrostatic precipitator Filtration - Theory of filtration, Filter equipment: Filter press, leaf filter, continuous rotary filter; filter medium, filter aid, filter auxiliaries; Centrifuges and centrifugal filtration. Project work on various types of filters used in industries Mixing of solids, liquids and slurries: Mixing mechanism, Scale up of Stirred vessels, Power consumption in stirred vessel, flow pattern in stirred tank, Mixing Equipment: Mechanical Agitator, Extruder, Static mixer. Pumping of fluids: Pumping equipment for liquids- Reciprocating pump, Rotary pumps, Centrifugal pump: Characteristic curves, Cavitation, NPSH, Priming; Pumping equipment for gases-Rotary blower and Compressors, centrifugal blowers & compressors, Power required for compression of gas, multistage compressors, compressor efficiency; Vacuum producing equipments Project work on industrially important pumps & their specific uses Material handling: storage of bulk solids- concept of sizing of bins; conveying of fluids, slurries and solids.		
List of Text Books/ Reference Books/Open sources	1. Unit Operations in Chemical Engineering by Warren L. McCabe, Julian C. Smith and C. Smith. 2. Principles of Unit Operations by Alan S. Foust, Leonard A. Wenzel and Curtis W. Clump 3. Unit Operations by George G. Brown 4. Unit Operations of Chemical Engineering: vol-1 Mechanical Operation by R. S. Hirenmath and A.P. Kulkarni		
Course Outcomes (COs):	The students of the course should be able to CO1:Analyze solid particle. (K4) CO2:Understand the procedure and importance of size reduction. (K2) CO3: Solve problems for motion of particles. (K3) CO4: Describe the procedures of solid handling. (K2) CO5: Explain basic concepts of filtration and pump. (K2)		

CO-PO Mapping

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3
CO1	3												3		
CO2	3	3											3		
CO3	3	3	3	3									3	3	
CO4	3	3	3										3	3	

Engineering Science	Course Code : Che/ES/B/Mech/T/226	MACHINE DESIGN	Credits: 3
	Total Contact Hours: 36		
Prerequisite			
	Course Content		**Hours**
	Basic idea of design, factor of safety, modes of failure, theories of failure, design under static and fatigue loading. Design of Cotter/knuckle Joint, threaded and riveted joint, eccentric loading. Shaft coupling (rigid / flexible). Belt-pulley drive. Pressure vessel.		
Course Outcomes (COs):	The students of the course should be able to CO1: Describe design philosophy, methodology and design considerations. (K2) CO2: Apply the concept of theories of failures and factor of safety to various engineering problems for static loading. (K3) CO3: Apply fatigue failure theories for simple and complex design problems. (K3) CO4: Describe the working principles of various mechanical joints and machine elements. (K2) CO5: Apply the basic concepts to design various mechanical joints and machine elements. (K3)		
CO-PO Mapping			

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3		1			**1**	**1**	**1**				
CO2	3	1	1									
CO3	3	2	1									
CO4	3	1	1									
CO5	3	2	2	1								

Sessional 4	Course Code : Che/ES/B/Mech/S/221	MACHINE DRAWING (Computer terminal mode)	Credits: 2
	Total Contact Hours:		
Prerequisite			
	Course Content		**Hours**
	Stuffing Box, Pipe Joints, Valves, Tool head of shaping machine, Plummer Block, Engine Parts.		
Course Outcomes (COs):	The students of the course should be able to – CO1: Describe the functions, uses and appropriate material of various machine components (K2) CO2: Demonstrate understanding of basic principles of engineering drawing and computer aided drafting (K3) CO3: Combine knowledge of engineering drawing and computer aided drafting to construct detailed part drawings of various machine components. (K5, S4) CO4: Combine knowledge of engineering drawing and computer aided drafting to accomplish complete assembly drawings of various machine components. (K5, S4)		
CO-PO Mapping			

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3					1	1					
CO2	3	1	1		3							
CO3	3	2	1		3							
CO4	3	2	1		3							

Sessional 5	**Course Code : Che/ES/B/Elec/S/222**	**ELECTRICAL ENGINEERING LAB**	**Credits: 1.5**
	Total Contact Hours:		
Prerequisite			
	Course Content		**Hours**
	Experiments to supplement the course on Basic Electrical Engineering (ES/EE/T101A).		
Course Outcomes (COs):	The students will be able to **CO1: Identify** the instruments required to perform a particular experiment (K1, S1) **CO2: Select** the ranges and ratings of the instruments identified (K2, S1) **CO3: Comprehend** the objective of the experiment and relate that with the acquired theoretical knowledge. **CO4: Realize** the electrical circuit duly connecting selected instruments and other apparatus (K2, S2) **CO5: Interpret** the experimental data and **prepare** a detailed report having graphs, charts etc. (K2)		
CO-PO Mapping			

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**	PSO1	PSO2	PSO3
CO1	3	2	1						2						
CO2	1	3	2						2						
CO3	1	3	2						2						
CO4	1	2	3						2						
CO5	1	1	2	4					2						

CHEMICAL ENGINEERING DEPARTMENT: BACHELOR OF CHEMICAL ENGINEERING

REVISED SYLLABUS OF BACHELOR OF CHEMICAL ENGINEERING

3RD YEAR 1ST SEMESTER

Program CORE	**Course Code: Che/PC/B/T/311**	**SEPARATION PROCESSES – I**	**Credits: 3**
	Total Contact Hours: 36		
Prerequisite			
	Course Content		**Hours**
	Diffusion: Molecular diffusion, Convective and diffusive mass transfer, Mass-transfer coefficients, relation between mass transfer coefficients, Mass transfer coefficient in laminar and turbulent flow, Theory of mass transfer (Film theory, Penetration theory, Surface renewal theory), simultaneous mass and heat transfer. Inter-phase mass transfer: Two-Film resistance theory, Local two-phase mass transfer, Local overall mass transfer coefficient; material balances for steady state concurrent and countercurrent processes- dilute solution approximation; stages, stage efficiency, Murphree stage efficiency; cross flow and countercurrent cascades. Equipment for Gas-Liquid Operations - Selection of equipment for mass transfer operation; Gas dispersed equipment; Operating characteristics of Tray towers, Point and Murphree tray efficiency; overall efficiency; Liquid Dispersed equipment; End effect and Axial mixing. Absorption (one component transferring, isothermal operation): equilibrium data representation; co-current operation and countercurrent operation, minimum liquid-gas ratio; countercurrent multistage operation; estimation of stages corrected for Murphree tray efficiency; Continuous contact: HETP, number and height of gas, liquid and overall transfer units, graphical construction of transfer unit. Humidification: Design of water cooling towers, Dehumidification of air-water vapour mixture, recirculating liquid-gas humidification-cooling, Equipment. Drying: Equilibrium data representation; Movement of moisture within the solid; Calculation of rate of drying in cross circulation drying, through circulation drying and continuous direct heat driers (high and low temperature); equipment. Project work on advanced drying technology		
List of Text Books/ Reference Books/Open sources	**1.** Mass Transfer Operations by Robert E. Treybal. 2. Principles of Unit Operations by ALAN S. FOUST, LEONARD A. WENZEL, CURTIS W. CLUMP, LOUIS MAUS, L. BRYCE ANDERSEN. 3. Unit Operations in Chemical Engineering by Warren L. McCabe, Julian C. Smith and Peter Harriott (7th Edition) 4. Separation Process Principles by Henley & Seader.		
Course Outcomes (COs):	From this particular course, students should be able to CO1:Characterize different principles of transfer Describing different models and Identification of different equipments based on different systems involved (A5, K2) CO2: Evaluate &Formulatevarious governing equations for different operational condition of mass transfer driven differential contactors (K6, K5) CO3: Apply theoretical background to Choose various mass transfer equipments as per applications (K3) CO4: Recognize themechanism of the steps involved in the processes to Describe the design steps for the equipments(K5, K2).		
CO-PO Mapping			

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	2	1								3	1	
CO2	3	3	2	2	3	-	-	-					3	1	
CO3	3	3	2	2	2								2	1	
CO4	3	1	2	-	-								2	2	

Program CORE	Course Code: Che/PC/B/T/312	CHEMICAL REACTION ENGINEERING-I	Credits: 4
	Total Contact Hours: 48		
Prerequisite			
	Course Content		**Hours**
	Kinetics of homogeneous reactions (*elementary and non-elementary*); autocatalytic Reactions; reversible reactions; interpretation of kinetic data; temperature and reaction rate. Mole Balances, Stoicheometry (*Batch and Flow processes*), Isothermal Reactor Design (*Batch, CSTR, PFR*), constant and variable volume reactions, Ideal reactors for a single reaction, Space time, Damkohler number, **Case studies on Thermal cracking of ethane, decomposition of nitrous oxide (N_2O)** Multiple reactor systems including reactor staging. Semi-Batch Reactors, Recycle reactors: performance and optimum recycle ratio. Multiple Reactions: design for irreversible series-parallel reactions; choice of reactors, selectivity and yield in specific combination of reactors. Non-isothermal reactors: Overview of energy balance equation; temperature and pressure effects on single and multiple reactions, relative rates; design considerations; adiabatic reactors: operating line and design considerations. C**ase study on adiabatic Isomerization of n-Butane** Reversible reactions in non-isothermal reactors: van't Hoff equation; equilibrium constant as a function of temperature; specific design considerations for exothermic and endothermic reactions; optimum temperature progression in various types of reactors; **Case study on γ-hydroxybutyric acid to form its lactone to determine 1st order Reversible reaction kinetics;** Reversible exothermic reaction with inter-stage and cold-shot cooling. Non-ideal reactors: Basics of non-ideal flow; contacting pattern; RTD: definitions of RTD functions; evaluation through tracer experiments (inputs: *step and pulse*); introduction to non-ideal reactor models: zero and one parameter models; calculation of conversion in non-ideal reactors.		
List of Text Books/ Reference Books/Open sources	1. Chemical Reaction Engineering by Octave Levenspiel (3rd Edition) 2. Elements of Chemical Reaction Engineering by H. Scott Fogler (4th Edition) 3. Chemical Engineering Kinetics, J. M. Smith (3rd Edition)		
Course Outcomes (COs):	**The students of the course should be able to** CO1 Define and describe kinetics of irreversible and reversible homogeneous reactions (elementary and non-elementary) and interpret kinetic data for isothermal Batch, CSTR, PFR using design equationK1 , K2 and K6 CO2 Determine performance of multiple flow reactor systems for constant and variable volume single CO3 Predict the efficacy of Semi-Batch Reactors, Recycle reactors and evaluate selectivity and yield for irreversible series-parallel reactions in different reactors K6 CO4 Formulate energy balance equation for adiabatic reactors: operating line and design considerations and optimum temperature progression in various reactions K5 CO 5 Assess non-ideal reactor behavior using RTD and employ zero and one parameter models to predict conversion K3 and K6		
CO-PO Mapping			

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3	3	3									
CO3	2	3	3	3								
CO4	2	3	3	3								
CO5	3	3	2	3								

Program CORE	Course Code: Che/PC/B/T/313	CHEMICAL TECHNOLOGY I	Credits: 3
	Total Contact Hours: 36		
Prerequisite			
	Course Content		**Hours**
	Water for the chemical process industry and its treatment: Boiler feed-water, Cooling tower water, Process Plant water, Treatment of water: lime-soda process, Flocculation, aeration, deaeration, ion-exchange. *Acid industries:* Sulfuric, hydrochloric and nitric. **Case studies on controlling tail gas emissions from acid industries** *Chlor-Alkali industries:* Caustic soda, Sodium carbonate, Chlorine, Bleaching powder. **Group discussion on "Evaporation and Handling of Caustic and Chlorine"** *Pulp and paper Paints and Varnishes. Cement.* **Case studies on abatement of dust pollution in cement industries** *Fertilizers:* Nitrogen fertilizers - synthetic ammonia, urea, ammonium chloride, CAN, ammonium sulphate. Phosphorous fertilizers - phosphate rock, phosphoric acid, superphosphate and triple super phosphate, MAP, DAP. Potassium fertilizers – potassium chloride and potassium sulphate.		
List of Text Books/ Reference Books/Open sources	1. Shreve's Chemical Process Industries by George T. Austin 2. Dryden's Outlines of Chemical Technology for the 21st Century 3rd Edition		
Course Outcomes (COs):	**Students will be able to** CO1: Understand and analyze the techniques used in treatment of water for chemical process industries. (K2) CO2: Able tosynthesis acids, alkalies, bleaching powder, paints, and varnishes. (K2, K3, K4) CO3: Able to analyze and synthesis cement. (K2) CO4: Able to synthesis nitrogen fertilizers, phosphorous fertilizers, and potassium fertilizers. (K2, K3)		
CO-PO Mapping			

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3
CO1	3												1		
CO2	3	3	3										2		
CO3	2	3	3	3									1		
CO4	2	3	3	3									1	3	
CO5	3	3	2	3									2		

Program CORE	Course Code: Che/PC/B/T/314	PROCESS HEAT TRANSFER	Credits: 3
	Total Contact Hours: 36		
Prerequisite			
	Course Content		**Hours**
	Introduction to Conduction, Convection and Radiation. Conduction: Mechanism of heat transfer, Fourier's law, heat transfer under steady state through composite slabs, composite cylinders and composite spheres. Specific case studies considering heat generation/depletion terms. Critical insulation thickness for composite cylinders and composite spheres. Unsteady state heat conduction: Concept of Lumped Parameters Analysis, introduction to Biot number and its applications, path of least thermal resistance. Case studies on sudden thermal quenching like i) cooling of a metallic wire in oil and in water, ii) transient temperature profile in a boiling egg, iii) performance of a thermocouple in terms of its response time, iv) unsteady-state conduction with convective boundary condition. Temperature changes with time and position. Use of Heisler's , Gurney Lurie & Similar type of Charts and their limitations. Case studies with specific geometries. Convection: Heat flow mechanism by convection. Individual & overall heat transfer coefficients, log-mean temperature difference [LMTD]. –Forced convection past a flat plate, inside and outside tubes and ducts - Derivation of a correlation between dependent and independent variables for convective heat transfer under turbulent flow through pipes using dimensional analysis, definition of forced convective heat transfer coefficient. Dittus-Boelter equation, considerations for viscosity correction. Reynold's analogy, Colburn analogy, Prandtl analogy. Various case studies on forced convective laminar and turbulent flows i) past a flat plate, ii) inside and outside the pipes using specific Nusselt Number correlations; Role of L/D ratio and use of charts. –Natural convection: Derivation of an expression for heat transfer in terms of the system variables for a plate at a particular temperature, dipped in a large volume of liquid at a constant temperature. Introduction to Grashof Number and Rayleigh Number. Considerations for specific Nusselt number correlations. Case studies past i) a flat vertical plate, ii) a horizontal plate and iii) a pipe. Heat Transfer with phase change: Condensation of single and mixed vapors: Dropwise and Filmwise condensation; Laminar Film condensation on a vertical wall- expressions for film thickness and specific Nusselt number correlation. Case study on Film condensation of saturated steam under laminar flow to find i) average heat transfer coefficient, (ii) total condensation rate, (iii) film thickness. Heat transfer to boiling liquids: Force balance on a submerged spherical bubble in a liquid; excess temperature and various regimes of pool boiling-natural convective, nucleate, transition and film. Insights of Peak Heat Flux and Leidenfrost point. Case study on burning out heat flux for nucleate boiling in a kettle boiler. Heat Exchangers: Co-Current and Counter-current flow-pattern, LMTD. Cross-flow pattern in multiple-pass tubular heat exchanger: LMTD Correction factor, effectiveness factor, NTU method. Types of heat exchangers: Shell-and-tube, Double-pipe, Compact [Plate-type], Condenser and Reboiler, Finned. Design of Shell and Tube heat exchangers: Fixed and floating heads, tube sheet, pitch, baffles, tie rods etc.. Various case studies on design of i) Shell-and-tube heat exchangers, ii) Double-pipe heat exchanger and iii) Condensers.		

	Radiation: Concept of black body and laws of black body radiation, Kirchhoff's law, emissivity, radiant heat transfer between surfaces separated by non-absorbing media, radiation error in Pyrometry. Radiative heat transfer coefficient. Radiation shield. Case studies on combined modes of heat transfer with special applications to furnaces and circular hot air ducts. Evaporation: The mechanism of vaporization, single & multiple effect evaporators, types of evaporators with accessories, various feeding arrangements, design calculations for evaporators and optimum number of effects, Thermo-compression evaporator.	
List of Text Books/ Reference Books/Open sources	1. Heat Transfer by J.P. Holman & S. Bhattacharya. 2. Process heat Transfer by Donald Q. Kern	
Course Outcomes (COs):	**Students will be able to** CO1 Understand the fundamentals of steady and un-steady state heat conduction and apply this knowledge to solve complex problems in various geometry related to conductive heat transfer (K2 and K3). CO2 Understand the basics of natural and forced convective heat transfer in order to solve problems related to internal and external flow conditions (K2 and K3). CO3 Understand the basics of radiative heat transfer, calculate view factor for simple and complex geometrics along with specific chemical engineering applications (K2 and K3). CO4 Understand the basics of heat transfer with phase change and solve problems for specific cases (K3). CO5 Design, compare and analyze double pipe and shell & tube heat exchanger performances. (K4 and K5) CO6 Understand mechanism of evaporation, formulate the balance equations and design single and multiple-effect evaporators (K2 & K5).	
CO-PO Mapping		

	PO1	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO1 0**	**PO1 1**	**PO1 2**	**PSO 1**	**PSO 2**	**PSO 3**
CO1	2	3	1										1		
CO2	3	2	1										2		
CO3	2	3	2										1		
CO4	3	2	1										1		
CO5	1	2	1										2	3	
CO6	1	2	1										2	3	

Program CORE (H)	Course Code: Che/PC/H/T/315	ENERGY ENGINEERING	Credits: 3
	Total Contact Hours: 36		
Prerequisite			
	Course Content		**Hours**
	Introduction -Survey of different sources of energy and their utilization. Fossil Fuels – Coal: characterization, coal beneficiation circuit; Petroleum based fuels and characterization:, Natural gas: Characterization, processing and distribution; Shale oil and gas Case studies on carbon capture and sequestration to reduce carbon footprintProcessed Fuels - Charcoal, coke, water gas, producer gas, refinery gas, LPG, CNG, LNG and synthetic petrol. Case studies on novel techniques of coal beneficiation of high ash Indian coals Nuclear Fuels - Sources of nuclear fuels and introduction to nuclear reactions. Solar Energy-photovoltaic cell, Effective utilization of solar energy for room heating, water heating and other industrial processes. Wind energy, geothermal energy and tidal energy. Energy from Biomass- bio-ethanol, biogas, bio-butanol, biodiesel, pyrolysis oil. Concept of biorefinery. Energy utilization - Thermodynamic and economic aspects of utilization. Burners, stokers, and furnaces for combustion of various conventional fuels. Waste Heat Recovery – Recuperators. Fuel cell - working principle, application. Case studies on use of oxy-fuel combustion/IGCC/co-generation/super-critical boilers for steam and electricity generation		
List of Text Books/ Reference Books/Open sources	1. Renewable Energy Resources: Basic Principles and Applications by G. N. Tiwari and M. K. Ghosal 2. Solar Energy: Principles of thermal Collection and Storage by S. P. Sukhatme 3. Elements of Fuels, Furnaces and Refractories by O. P. Gupta 4. Nuclear Reactor Engineering by Samual Glasstone and Alexander Sesonske 5. Handbook of Natural Gas Engineering, D. L. Katz. 6. Thermochemical Processing of Biomass, A. V. Bridgwater		
Course Outcomes (COs):	Students will be able to CO1: Categorize different sources of energy and their utilization (K5 & K4). CO2: Understand and analyze the process of generation of solid fuel and gaseous fuel (K2 & K4). CO3: Differentiate and compare the process of generation of various liquid fuels (K2 & K6). CO4: Distinguish and illustrate different process and mechanism related to nuclear fuel and non-conventional fuels (K2 & K3).		
CO-PO Mapping			

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3						3				1		1		2
CO2	2	3			3		3							2	2
CO3	2	3	3		3		3							2	
CO4	2	3	3	1	2		1						1		2

Sessional 6	Course Code : Che/PC/B/S/311	MOMENTUM TRANSFER AND MECHANICAL OPERATION LABORATORY	Credits: 2
	Total Contact Hours:		
Prerequisite			
	Course Content		**Hours**
	Study of pipe fittings, valves and other equipment, pressure drop for flow through pipe line -valves and fittings, Pump characteristics, Flow measurement by orifice meter, venturimeter, rotameter, pitot tube, weirs. Pressure drop for flow through packed bed and fluidised bed, studies on crushing and grinding, screening efficiency. Studies on filtration and other mechanical operation.		
Course Outcomes (COs):	CO1: interpret the experimental results based on theoretical knowledge of Momentum Transfer & Mechanical Operation (K2, K3) CO2: Demonstrate hands on experience of the operation of the equipment (K4) CO3: Perform experiments coherently in a group (K6)		
CO-PO Mapping			

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	**3**	**3**	**2**	**2**	**2**			**2**	**2**	**2**		**2**	**3**	**2**	
CO2					**2**				**2**			**2**	**2**	**3**	
CO3								**2**	**3**	**3**		**2**	**3**	**2**	

Sessional 7	Course Code : Che/PC/B/S/312	COMPUTER APPLICATIONS IN CHEMICAL ENGINEERING LABORATORY	Credits: 2
	Total Contact Hours:		
Prerequisite			
	Course Content		**Hours**
	Solution of chemical engineering problems using FORTRAN, C/C++, MATLAB, CHEMCAD or HYSIS		
Course Outcomes (COs):	**Students will be able to** CO1 numerically solve chemical engineering problems CO2 use C/C++/MATLAB as a tool for solving complex chemical engineering problems CO3 address complex process engineering problems using commercial softwares such as ASPEN10, UNISIM.		
CO-PO Mapping			

	PO1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO 2	PSO 3
CO1	3	3	3						3				1	2	
CO2	2	3	3	3					2				3	1	2
CO3	3	3	3	3					3				3	3	3

CHEMICAL ENGINEERING DEPARTMENT: BACHELOR OF CHEMICAL ENGINEERING

REVISED SYLLABUS OF BACHELOR OF CHEMICAL ENGINEERING

3RD YEAR 2ND SEMESTER

Program CORE	**Course Code: Che/PC/B/Elec/T/321**	**PRINCIPLES OF MEASUREMENT AND INSTRUMENTATION**	**Credits:**
	Total Contact Hours: 36		
Prerequisite			
	Course Content		**Hours**
	Introduction: Need for measurement of Process variables, Classification of measurement errors (Deterministic and Probabilistic) Steady state and Dynamic characteristics of instruments, Instrument calibration technique, Selection of instruments. Process transducers and application: Temperature Measurement - Thermocouple, RTD, Optical & Radiation pyrometer. Pressure measurement- manometers, instruments based on mechanical elastic element, strain gauge, electrical transducers, High vacuum and differential pressure measurement. Flow measurement-Constant and variable area flow meters, Anemometers, Open channel flow - Weirs, Special flow measurement devices, Granular solid flow measurement. Level measurement - Direct and Inferential type instruments, Granular solid level measurement. Chemical composition measurement based on - Optical properties, Thermal conductivity, para-magnetic properties, conductivity of solution, Combustion products, adsorption properties, Chomatography, Molecular and Atomic Properties. Density, Viscosity and Humidity measurement. Instrumentation Devices and Systems: Controller - Pneumatic, electronic, analog and digital controller, self-tuning and multifunction controllers. Actuators: Pneumatic, Hydraulic, Valve positioner. Indicating, Recording and Data acquition devices. Final controlling elements - Control valve and their types and characteristics, Valve sizing. Industrial Instrumentation Systems: Components, structure, specification, planning, monitoring and design, case study.		
List of Text Books/ Reference Books/Open sources	• Measurement Systems – Application and Design: Doeblin • Transducers and Instrumentation: D. V. S. Murty • Instrument Transducers: Neubert • Principle of Industrial Instrumentation: Patranabis • A Course in Electrical & Electronic Measurements & Instrumentation: A. K. Sawhney • Principles of Process Control: Patranabis • Chemical Process Control: Stephanopoulos • Process Control Engineering: P. Sai Krishna		
Course Outcomes (COs):	**CO1: Describe** the operating principles of various transducers. (K1) CO2: **Discuss** characteristics and errors of different types of instruments. (K2) CO3: **Comprehend** various techniques of measurement using transducers. (K3) CO4: **Design** appropriate controllers for their application in industrial processes. (K4)		
CO-PO Mapping			

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1										
CO2	2	2										
CO3	1	3	2	1								
CO4	1	3	2	1								

Program CORE	Course Code: Che/PC/B/T/322	PROCESS DYNAMICS AND CONTROL	Credits: 3
	Total Contact Hours: 36		
Prerequisite			
	Course Content		**Hours**
	Introduction to dynamics of processes - dynamic characteristics as an extension of steady state behavior - examples, dynamic model of a process; SISO, MIMO and MISO processes – examples. Properties of Linear systems. Concept of deviation variables. Application of Laplace Transforms for dynamic characterization of processes – concept of Transfer function. Properties of transfer functions – additivity and multiplicity. Definition and characterization of some standard inputs/ forcing functions viz. step, ramp, pulse – rectangular and trapezoidal, impulse, sinusoidal, exponential etc. Characterization of First order, Pure Integrator and Second order (Interacting and non-interacting) systems in time domain and Laplace domain – standard forms of corresponding ODEs and Transfer Functions. Dynamic Responses of First Order, Second Order and Pure Integrator systems to some standard input/ forcing functions. Nonlinearities in chemical process models, examples – linearization. Responses of linearized systems and comparison to actual nonlinear responses by numerical simulation. Systems with dead time – non-polynomial transfer functions and use of Pade approximation for polynomial approximations to such transfer functions. First-order plus dead time (FOPDT) systems. Stability – concept of BIBO stability. Pole-zero characterization of transfer functions of processes - location and movement of poles on a complex s-plane and impact on stability. Transfer functions of processes with numerator dynamics. Concept of feedback control of linear systems – control systems and their constituent elements, block diagram based representation, open-loop and closed-loop dynamics of controlled processes; effect of recycling on dynamics of chemical processes – examples. Controllers – proportional, integral and derivative modes of control action - the PID Control Algorithm, P, PI and PID controllers – their dynamic characteristics and effects on the dynamics of controlled processes; Uses and applications of various types of controllers in the Chemical Process industry through industry-oriented project work. Stability of dynamic systems - Routh-Hurwitz criteria; Root Locus method. Selection and Tuning of Controllers - Methods based on FOPDT model and Process Reaction Curve (PRC); Integral Error Criteria; Direct Substitution Method; Model-independent methods based on field trials - Continuous Cycling and Frequency Response. Bode Stability Criterion. Nyquist Stability Criterion; Ziegler-Nichols and Cohen-Coon optimum controller settings. Development of algorithms and /or softwares for real time stability analysis and controller tuning of process control systems. Advanced Control Strategies – (Preliminary treatment only)- Ratio Control, Feed forward control, Cascade control, Selective Override control, Adaptive control. Computer Based Control.		
List of Text Books/ Reference Books/Open sources	1. Process Dynamics and Control by Del E Seborg, Thomas F. Edgar and Duncan A. 2. Chemical Process Control: An Introduction to Theory and Practice by George Stephanopoulos		
Course Outcomes (COs):	**On successful completion of this course students will be able to** CO1: Understand & Analyze dynamic models of chemical processes based, fundamental principles of classical control theory, including different types of controllers and control strategies (K2, K4, K5) CO2: Evaluate & Formulate various governing equations for different operational condition using differential systems using Laplace Transforms, (K2, K3, K5, K6) CO3: Apply theoretical background and Choose loops using different controllers, stability of these controller systems and models developed, derive transfer function models from process models and process data (K2, K3)		

	CO4: Recognize the mechanism of the steps involved in the processes, apply modern control theory to design a controller, and Describe the design methodologies of the equipments(K5, K2)
CO-PO Mapping	

CO/PO	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO10**	**PO11**	**PO12**	**PSO 1**	**PSO 2**	**PSO 3**
CO1	3	3	3	2	1								3	1	
CO2	3	3	2	2	3	-	-	-					3	1	
CO3	3	3	2	2	2								2	1	
CO4	3	1	2	-	-								2	2	

Program CORE	Course Code: Che/PC/B/T/323	SEPARATION PROCESSES - II	Credits: 3
	Total Contact Hours: 36		
Prerequisite			
	Course Content		**Hours**
	Distillation: vapor-liquid equilibrium; steam distillation; single stage batch and flash distillation; multistage fractionating column (single feed, multi-feed and side stream) using McCabe Thiele & Ponchon-Savarit method; Azeotropic, extractive; Molecular distillation. **Project work on special distillation processes** Liquid-liquid Extraction: Equilibrium data representation; Method of calculation for single stage, Multistage Crosscurrent and Counter-current operation for one component transferring and all three component transferring; Equipment. Adsorption: Equilibrium data representation; Method of calculation for Single stage, multistage crosscurrent, multistage countercurrent adsorber(one and two component transferring); Fixed bed Adsorber - Break-through curves, Determination of time required to reach breakthrough point, length of unused bed; chromatography; Equipment. **Case study on adsorption technology for waste water treatment** Leaching: Mass transfer in leaching operation, Equilibrium data representation; Method of calculation for Single stage, Multistage Crosscurrent, Multistage Countercurrent Operation; Equipment. Crystallization: Equilibrium data representation, mechanism of crystallization, Estimation of crystal size distribution (CSD) in idealized crystallizers (MSMPR), equipment.		
List of Text Books/ Reference Books/Open sources	1. Mass Transfer Operations by Rober E. Treybal 2. Principles of Unit Operations by Alan S. Foust, Leonard A. Wenzel and Curtis W. Clump 3. Unit Operations in Chemical Engineering by Warren L. McCabe, Julian C. Smith and C. Smith		
Course Outcomes (COs):	From this particular course, students should be able to CO1:Understand&Analyze the equilibrium diagrams and Formulate mass balance and/or energy balance equations to solve staged, mass transfer based separation processes (K2, K4, K5) CO2: Evaluate &Formulatevarious governing equations for different operational condition of mass transfer driven separation processes (K2, K3, K5, K6) CO3: Apply theoretical background and Choose various mass transfer equipments as per applications (K2, K3) CO4: Recognize themechanism of the steps involved in the processes and Describe the design methodologies of the equipments(K5, K2)		
CO-PO Mapping			

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	2	1								3	1	
CO2	3	3	2	2	3	-	-	-					3	1	
CO3	3	3	2	2	2								2	1	
CO4	3	1	2	-	-								2	2	

Program CORE	Course Code: Che/PC/B/T/324	CHEMICAL TECHNOLOGY II	Credits: 3
	Total Contact Hours: 36		
Prerequisite			
	Course Content		**Hours**
	Petroleum Refining Operations: NG and LPG, Principles and details of Crude Distillation, Vacuum Distillation, Coking, Cracking, Catalytic reforming, Hydrotreatment and Merox. Petrochemicals: Raw materials and principles involved in the production of petrochemical precursors - olefins and aromatics, Butadiene and typical intermediates from olefins and aromatics such as ethylene glycol, ethyl benzene, phenol, cumene.Group discussion on balanced process for vinyl chloride production.Group discussion on reactor selection for alkylation of benzene with ethylene; Group discussion on recycle scheme of hydrogen stream in toluene hydrodealkylation process Case studies on recovery of flare gas from main flare system to minimize hydrocarbon loss in refinery and petrochemical complex Polymer industries: Plastics, rubbers, fibres, production of thermoplastic and thermosetting materials such as polyethylene, polypropylene, PVC, polyesters, phenolic resins and epoxy resins, natural and synthetic rubbers. Cellulose and cellulose products. Synthetic detergents Coal based chemical industries Oils and fats: Fat-splitting, refining, bleaching, deodorization, applications, e.g. soap manufacture, glycerine manufacture Sugar, Industrial fermentation and ethyl alcohol Leather		
List of Text Books/ Reference Books/Open sources	**1. Shreve's Chemical Process Industries by George T. Austin** **2. Dryden's Outlines of Chemical Technology for the 21st Century 3rd Edition** **3. Modern Petroleum Refining Processes by Bhaskar Rao, B. K.** **4. Petrochemical technology by Bhaskar Rao.**		
Course Outcomes (COs):	Students will be able to CO1: Understand the principles and operation of natural gas, liquefied petroleum gas, and crudes distillation. (K2 and K3) CO2: Able to learn the operation of coking, cracking, catalytic reforming, hydrotreatment, and Merox sweetening processes. (K2 and K3) CO3: Able to synthesize petrochemicals such as olefins and aromatics, butadiene, ethylene glycol, ethyl benzene, phenol, cumene. (K2 and K3) CO4: Understand the production of different polymers such as Plastics, rubbers, fibres, polyethylene, polypropylene, PVC, and polyesters. (K3) CO5: Able to learn the synthesis of detergents and soaps. (K4 and K5) CO6: Understand the operation method of industrial fermentation and leather synthesis. (K2 & K5)		
CO-PO Mapping			

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO 2	PSO 3
CO1	2	3		1									1		
CO2	3	2	3	1									2		
CO3	2	3	3	2									1		
CO4	3	2	3	1									1		
CO5	1	2	2	1									2	3	
CO6	1	2	3	1									2	3	

Program CORE	Course Code: Che/PC/B/T/325	Mathematical Modeling in Chemical Engineering	Credits: 3
	Total Contact Hours: 36		
Prerequisite			
	Course Content		**Hours**
	Introduction: Mathematical Modeling of a process, Organization of equations. Development of Information flow diagram based on macroscopic mass, energy and momentum balance: Case studies on i) Heating of a solution inside a well stirred tank, ii) Temperature response of a slurry tank, iii) Analysis of piping network, iv) Consecutive reversible reaction in a constant volume batch reactor. Steady State Lumped system: Algebraic equations, Structural analysis and functionality matrix, determination of design variables. Unsteady state Lumped system: ODEs, Stiff ODEs. Thermodynamic and Mass transfer Operations: Case studies on i) Boiling in jacketed vessel, ii) Batch distillation, iii) Plate type absorption column, iv) Binary distillation column Heat Transfer Operations: Case studies on i) evaporation of a sugar solution in an evaporator Mass transfer including reactions: Stirred Tank Reactor Modeling with case study on temperature effects in a stirred tank. Distributed parameter systems: microscopic balance, Heat conduction with chemical heat source. Case Study on Mathematical Modeling of a Tubular reactor for producing Styrene Monomer. Split Boundary Value Problem for Distributed systems- Case studies on i) Tubular reactor with dispersion, ii) Annular Chemical Reactor, iii) Countercurrent heat exchanger. Stability analysis: multiple steady states, bifurcation theory and applications. Group discussion on existence of multiple steady states. Case Study on Laboratory scale nonisothermal CSTR for decomposition of Hydrogen peroxide (Nyquist and Ramirez, 1971): Development of model equations, material and energy balance, phase plane analysis with and without control, stability analysis, simulation. Case Study on modeling and simulation of Polystyrene Reactor.		
List of Text Books/ Reference Books/Open sources	1. Computational Methods in Process Simulation by W. Fred Ramirez 2. Mathematical Methods in Chemical Engineering by V. G. Jenson and G.V. Jeffreys 3. Mathematical Methods in Chemical Engineering: Matrices and Their Application by N. R. Amundson 4. Mathematical Methods in Chemical Engineering by S. Pushpavanam 5. Mathematical Methods in Chemical Engineering by A. Verma and M. Morbidelli 6. Applied Numerical Methods with Personal Computer by A. Constantinides		
Course Outcomes (COs):	**The students of the course should be able to** CO1 Define and describe mathematical modeling of Chemical Engineering systems K1 and K2 CO2 Develop macroscopic balance equations and Formulate models for steady and unsteady lumped systems K3 and k5 CO3 Develop microscopic balance equations and Formulate models for distributed systems K3&K5 CO4 Solve model equations and analyze lumped and distributed systems k4 CO 5 Explainstability analysis with emphasis on multiplicity and bifurcation K6		

CO-PO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	2	3	3	3								
CO3	2	3	3	3								
CO4	3	3	3	2								
CO5	3	3	2	3								

Program CORE	Course Code: Che/PC/B/T/326	CHEMICAL REACTION ENGINEERING - II	Credits: 3
	Total Contact Hours: 36		
Prerequisite			
	Course Content		**Hours**
	Non-catalytic Heterogeneous Reactions: Selection of model: Progressive Conversion model and Unreacted core model; Determination of controlling step, Application to design.		**5**
	Catalysis and Catalytic Reactors: overview of solid catalyzed reactions: rate equations for surface kinetics; mass transfer between the bulk fluid phase and external catalyst surface in isothermal reactors; pore and film diffusion resistances; deactivation and regeneration of catalysts. Group discussion on the deactivation of catalyst with time with the help of a case study on the cracking of gas oil in a fluidized bed reactor; Group discussion on the deactivation of catalyst with time using a case study on the cracking of gas oil in a moving bed reactor.		**10**
	Porous catalyst particles: Deriving the global reaction rate expression; determination of rate controlling step; effectiveness factor for flat-plate, cylindrical and spherical catalyst pellets.Group discussion on the determination of overall effectiveness factor using the case study on the reduction of nitrous oxide in plant effluent; Group discussion on the determination of rate limiting step;case studies on copper catalyzed surface reaction between NO and CO.		**7**
	Performance equation for catalytic reactors with porous catalysts; Case study on catalytic hydrodealkylation of toluene for synthesis of benzene using differential reactor data;Case study of hydrogenation of oil in slurry reactor and in trickle bed reactor;pressure drops in packed bed catalytic reactors; Heat effects in catalytic reactors; adiabatic packed bed catalytic reactors.		**8**
	Biochemical reaction kinetics and bioreactors: Enzymatic Reaction: Michaelis-Menten Kinetics; Competitive and Non-competitive Inhibition.		**3**
	Microbial Reaction: Microbial growth kinetics (Monod); substrate and product limiting scenarios for Batch, plug and Mixed flow, optimum operation of bioreactors. Multiplicity: Steady state multiplicity in CSTRs.		**3**
List of Text Books/ Reference Books/Open sources	1. Chemical Reaction Engineering by O. Levenspiel (3rd Edition) 2. Chemical Engineering Kinetics by J. M. Smith (3rd Edition) 3. Elements of Chemical Reaction Engineering by H. S. Fogler 4. 4th Edition Reaction Kinetics for Chemical Engineers by S. M. Walas 5. Chemical Reactor Analysis and Design by K. B. Bishop and G. F. Froment		
Course Outcomes (COs):	**The students of the course should be able to** CO1 Define and describe the basic mechanisms of non-catalytic and catalytic heterogeneous reactions and biochemical reactions K1 and K2 CO2 Develop rate equations for different types of heterogeneous reactions and biochemical reactions K3 CO3 Determine the controlling steps for heterogeneous reactions K4 CO4 Formulate design equations for heterogeneous reactors and bioreactors and predict their performance K5& K6 CO 5 Explain steady state multiplicity in CSTRs K6		

CO-PO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3	3	3									
CO3	2	3	3	3								
CO4	2	3	3	3								
CO5	3	3	2	3								

Sessional 8	Course Code: Che/PC/B/S/321	REACTION ENGINEERING AND THERMODYNAMICS LAB	Credits: 2
	Total Contact Hours:		
	Course Content		**Hours**
	Introduction to all reactors of the Reaction Engineering and Thermodynamics Laboratory		
	Experiments		
	1.Kinetics and Performance study of a stirred batch reactor.		
	2.Kinetics and Performance study of an isothermal Continuous Stirred Tank Reactor (CSTR)		
	3.Kinetics and Performance study of an isothermal Semibatch Reactor		
	4.Kinetics and Performance study of an isothermal Plug Flow Reactor (Straight Tube type)		
	5.Kinetics and Performance study of an isothermal Packed Bed Reactor		
	6.Performance study of isothermal Cascade CSTRs		
	7.RTD Experiment using Step type tracer input for the determination of non-ideality of a CSTR using two parameter model		
	8.RTD Experiment using Step type tracer input for the determination of non-ideality of a Plug Flow Reactor using dispersion Model		
	9.RTD Experiment using Pulse type tracer input for the determination of non-ideality of a Plug Flow Tubular Reactor (Straight Tube type)		
	10.Study of the performance of an adiabatic batch reactor		
	11. Study of the performance of a Plug flow reactor with recycle		
Prerequisite	Theoretical knowledge on homogeneous reaction kinetics and Reactor operation		
Course Outcomes (COs):	The students of the course should be able to CO1 Examine the behavior of batch, semi-batch, single and multi-stage continuous stirred tank reactor (CSTR), Plug flow reactor (PFR), and packed bed reactor (PBR). K1,A2 CO2 Characterize the behavior of batch and semi-batch reactors and collect data to determine the rate kinetics of homogeneous reactions A2, A5, K4 CO3 Analysethe behavior of idealCSTR , PFR and PBR by scheduling experiments to evaluate rate equation A2, K3, K4, CO4 Evaluate the non-ideality of CSTR and PFR by performing RTD experiments K4, K6 CO 5 Assessthe behavior of multi-stageCSTRs in series and compare the performance with single CSTR A2, K6		

CO-PO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3			3					3			
CO2	3	3	3	3					3			
CO3	3	3	3	3					3			
CO4	3	3	3	3					3			
CO5	3	3	3	3					3			

Sessional 9	Course Code: Che/PC/B/S/322	ENERGY ENGINEERING LAB	Credits: 1.5
	Total Contact Hours:		
Prerequisite			
	Course Content		**Hours**
	Proximate analysis of coal		
	Properties of liquid fuels: Flash point & Fire point, Carbon residue, Viscosity & Viscosity Index, Distillation characteristics.		
	Hard groove grindability index of coal.		
	Calorific values of solid, liquid & gaseous fuels.		
	Ors at analysis of gaseous mixture.		
	Washability of coal		
	Caking Index of coal, Swelling Index of coal.		
	Studies of operating character of solar collectors.		
	Studies on flame stability and burner characteristics.		
	Studies on properties of bio-mass.		
Course Outcomes (COs):	**The students will be able to** CO1 interpret the experimental results based on theoretical knowledge of energy engineering and chemical technology CO2 demonstrate hands on experience of different experiments related to fuel characteristics CO3 perform experiments coherently in a group		
CO-PO Mapping			

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PS O 2	PS O 3
CO1	3	3	2	2									3	3	3
CO2	3	3	2	3									3	2	3
CO3	2	3	2	3									3	2	3

Sessional 10	Course Code: Che/PS/B/S/313	CHEMICAL ENGINEERING PROJECT - I	Credits: 2
	Total Contact Hours:		
Prerequisite			
	Course Content		**Hours**
	Specific projects/designs will be assigned by the teachers concerned		
Course Outcomes (COs):	**Students will be able to** CO1apply chemical engineering knowledge to address a specific research /design problem CO2 assimilate the existing knowledge base for specific research /design problem CO3 incorporate new ideas to create originality in the specific research /design problem CO4 write report based on experimental results/ design solutions for the specific research /design problem.		
CO-PO Mapping			

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PS O 2	PS O 3
CO1	3	3	3	3	3						3		3	3	3
CO2	3	3	3	3									3	3	3
CO3	3	3	2	3									1	3	3
CO4	3	3	1	2	3				3			3	1	1	3

CHEMICAL ENGINEERING DEPARTMENT: BACHELOR OF CHEMICAL ENGINEERING

4TH YEAR 1ST SEMESTER

Program Honors	**Course Code: Che/PC/H/T/411**	**SEPARATION PROCESSES - III**	**Credits: 3**
	Total Contact Hours: 36		
Prerequisite			
	Course Content		**Hours**
	Specification of design variables: Stream variables, Adiabatic equilibrium stage, Equilibrium stage with heat addition, feed stream and side stream, Condenser and boiler, Combinations of elements by an enumeration algorithm Multicomponent flash calculation: Isothermal flash calculation, Adiabatic flash calculation Approximate methods for multicomponent – multistage separation: Design methods and simulation methods for multistage contactor, Fenske – Underwood – Gilliland (FUG) method for distillation, Multistage counter-current cascade – Group method for absorber and stripper, Kremser method, Horton and Franklin method, Edmister method (without derivation) Rigorous method for multicomponent – multistage separation: Introduction to MESH equation, Classification of different method based on solution scheme, **Project work on** Thiele – Geddes method with theta (θ) method of convergence, Membrane Separation Process: Different types of membranes, module design, concentration polarization and other effects, characterization of membranes, process design Classifications of different membrane separation processes: Pressure-driven membrane processes - microfiltration, ultrafiltration, nanofiltration and reverse Osmosis. **Case study on desalination of water.** Other membrane separation processes - Dialysis, Electrodialysis, Pervaporation Liquid membrane: Emulsion Liquid membrane, supported liquid membrane, New membrane separation processes like Membrane Distillation, Membrane Contactor and hybrid processes.		
List of Text Books/ Reference Books/Open sources	1. Mass Transfer Operations by Rober E. Treybal 2. Transport Processes & Unit Operations by Chrislie J. Geankoplis. 3. Unit Operations in Chemical Engineering by Warren L. McCabe, Julian C. Smith and C. Smith 4. Membrane Handbook by W. Ho & K. Sirkar. 5. Equilibrium-Stage Separation Operations in Chemical Engineering by Ernest J. Henley, J. D. Seader		
Course Outcomes (COs):	Students will be able to CO1. To understand and analyze the fundamentals of steady state multicomponent–multistage separation (K2 and K4). CO2. To developthe design equations for various types of multicomponent–multistage contactors using both approximate and rigorous methods and apply the same to solve practical problems (K3, K5 and K6). CO3. To understandandanalyzethe basics of membrane separation processes highlighting mechanism of specific processes, module types and characterization of membranes. (K2 and K4) CO4. Design, compare and analyze various traditional and state-of-the-art membrane separation processes. (K4, K5 and K6)		
CO-PO Mapping			

	PO 1	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO1 0**	**PO1 1**	**PO1 2**	**PSO 1**	**PSO 2**	**PSO 3**
CO1	3	2											3		
CO2		3	3		2									3	
CO3	3	3					2						3		
CO4	2	2	3										3	3	

Program Honors	Course Code: Che/PC/H/T/412	OPTIMIZATION METHODS IN CHEMICAL ENGINEERING	Credits: 4
	Total Contact Hours: 48		
Prerequisite			
	Course Content		**Hours**
	Introduction to optimisation; Basic concepts; Formulation of objective function; functions, regions, necessary and sufficient conditions for an extremum of an unconstrained function. One dimensional Search: Scanning and bracketing; Newton, quasi-Newton and secant methods, Region elimination method, Polynomial approximation methods. Unconstrained multivariable optimization: Direct methods-random search, grid search, univariate search, simplex method, conjugate search direction and Powell's method. Indirect method- gradient and conjugate gradient methods, Newton's method, movement in search direction, secant method. Linear programming: Basic concepts in linear programming; Graphical solution; Simplex method; Standard LP from obtaining first feasible solution. Non-linear programming: Lagrange multiplier method; Quadratic programming; Penalty function and augmented Lagrangian methods; Successive quadratic programming; Optimization of dynamic processes. Optimization of staged and discrete processes: Dynamic programming; Integer and mixed integer programming. Application in design of separation process, chemical reactor and process plant.		
List of Text Books/ Reference Books/Open sources	1. Optimization for Engineering Design – Algorithms and Examples by Kalyanmoy Deb 2. Optimization of Chemical Processes by Thomas F. Edgar, David M. Himmelblau, Leon Lasdon		
Course Outcomes (COs):	**Students will be able to** CO1: Understand and describe key concepts and techniques used in optimization. (K2, K1) CO2: Determine the parameter to be optimized, develop the problems as mathematical function, evaluate, and solve the model. (K3, K4, K5) CO3: Apply appropriate techniques and algorithms for solving an optimization problem and interpret and analyze the solution obtained thereof (K2, K3, K6) CO4: Solve and explain optimization problem associated to Chemical Engineering using software tools (K4, K6).		
CO-PO Mapping			

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	3	3	2									2		3
CO2	3	3	3	3									3		3
CO3	3	3	3	3	2								3	2	3
CO4	3	3	3	3	3	2	2				2	2	3	2	3

Program Honors	Course Code: Che/PC/H/T/413	INDUSTRIAL POLLUTION CONTROL ENGINEERING	Credits: 3
	Total Contact Hours: 36		
Prerequisite			
	Course Content		**Hours**
	Definition of pollution Air Pollution: Types of Pollutants, Sources of Pollutants, Classification of Pollutants: particulates, gaseous, hydrocarbons, CO, SO_x, NO_x, Criteria pollutants, GHGs and global warming, Control Devices for Particulate Contaminants: Gravitational Settling Chambers, Impact separator, centrifugal Collectors, Wet Collectors, Fabric Filters, Electrostatic Precipitators, Control Devices for gaseous Contaminants: Adsorption, Absorption, Venturi scrubber, Bio-scrubbers. Water Pollution: Introduction, Water quality standards and parameters, Types of pollution, Wastewater Treatment: Pre-treatment of water—screening, degritting, filtration, coagulation and flocculation; Primary Treatment -- primary sedimentation and clarification; Secondary Treatment- types of microorganism, growth and food utilization, activated sludge process, biotrickling filters, ponds and lagoons, secondary clarification; Tertiary Treatment-- membrane separation processes, ion exchange. Solid waste management: Landfilling, Bioremediation of soil, Waste to energy concept, incineration, Hazardous Waste Treatment: introduction, definition of hazardous waste, hazardous waste treatment facility, disposal of hazardous waste. Anaerobic digestion: microbiology of anaerobic digestion, sizing of anaerobic digesters, characteristics of methane production. A case study on air pollution from the Industrial chimney, Group discussion on suitable disposal of hazardous waste.		
List of Text Books/ Reference Books/Open sources	1. Wastewater Engineering: Treatment and Reuse, Metcalf & Eddy, George Tchobanoglous, Franklin Burton , Mc Graw Hill. 2. Environmental Pollution Control Engineering, C S Rao, New Age International Publishing 3. Physical-Chemical Treatment of Water and Wastewater, Arcadio P Sincero , Gregoria A. Sincero , CRC Press		
Course Outcomes (COs):	The students of the course should be able to CO1: Understand the different types of pollutants present in air, water, industrial water and its analysis. K2 CO2: Able to solve numerical problems and design control devices to remediate water and air pollution K2, K3, K4 CO3: Able to classify hazardous and nonhazardous solid waste wastes, treatment and disposal. K2 CO4: Able to understand various remediation processes and its application to control pollution K2, K3		
CO-PO Mapping			

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO11 0	PO11 1	PO1 2
CO1	3	3	3	2			2					
CO2	3	3	3	3			2					
CO3	2	3	3	2	2	2	3					
CO4	3	2	2	1	3	1	2					

Program Elective	Course Code: Che/PE/B/T/414A	INTERFACIAL SCIENCE AND ENGINEERING (ELECTIVE – I)	Credits: 3
	Total Contact Hours: 36		
Prerequisite			
	Course Content		**Hours**
	Physics of surfaces Surface and Interface- molecular origin, the work of cohesion and adhesion Interaction forces and potential, chemical and physical interaction, classification of physical forces. Van der Waals force - **Case studies on spherical particles, sphere and half space, parallel cylinders**, interaction between surface and particles – Lifshitz theory, hydrodynamic flow effects in interfacial interactions. Electrostatics forces and electric double layer, DLVO theory, Boltzmann distribution, Debye length- **Group discussion on effect of different types of salts(effect of valency)**, specific ion adsorption, ion adsorption, Stern layer, Electrphoresis Theory of capillarity – Young Laplace equation-**Case studies on effect of different geometries**, **Case study on soap film stretched between two circular parallel ring**, capillary flow and spreading process, wetting and spreading, contact angle and its measurement technique, contact angle hysterisis, Thermodynamics of wetting, Young's equation, spreading coefficient Equilibrium dynamics and stability of free liquid surfaces Chemical equilibrium across a curved interface, generalized Kelvin equation. Oswald ripening, Capillary condensation-**Group discussion on effect of different geometries**, Theory of heterogeneous nucleation Effects of solutes on surface tension-**A case study on effect of surfactant on water strider's walk on water**, Adsorption, Gibbs surface excess, Gibbs adsorption equation, Adsorption isotherms, surfactant adsorption Surfactants, self assembly thermodynamics, bilayers, vesicles, phase diagram Stability of colloids – emulsions, microemulsions, foams, aerosols, polymers at Interface, Application involving various colloidal systems. **A case study on self-cleaning coatings**. Nanotechnology: Nanostructured materials, Nanofabrication and material characterizations. **Group discussion on top-down and bottom-up approaches**.		
List of Text Books/ Reference Books/Open sources	1. Mass Transfer Operations by Rober E. Treybal 2. Transport Processes & Unit Operations by Chrislie J. Geankoplis. 3. Unit Operations in Chemical Engineering by Warren L. McCabe, Julian C. Smith and C. Smith 4. Membrane Handbook by W. Ho & K. Sirkar. 5. Equilibrium-Stage Separation Operations in Chemical Engineering by Ernest J. Henley, J. D. Seader		
Course Outcomes (COs):	Students will be able to CO1. To understand and describe fundamentals of colloid and interface science and their application in real life and in industry. K1, K2. CO2. To calculate and analyze various inter particle forces for colloidal systems. K2, K3 CO3. To develop the governing equation for an interface by applying the concept of surface tension, capillarity and to manipulate wettability of surfaces. K3, K5 CO4. To explain the role of surfactants in stability of colloids for specific cases. K4 CO5. To describe and apply the techniques of nanomaterial fabrication and characterization. K2, K4, K5		
CO-PO Mapping			

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2					2								
CO2	2	3	2												
CO3	2	3	1												
CO4	3		2												
CO5	2			3											2

Program Eletive	Course Code: Che/PE/B/T/414B	MULTIPHASE FLOW (ELECTIVE – I)	Credits: 3
	Total Contact Hours: 36		
Prerequisite			
	Course Content		**Hours**
	Importance of multiphase flow. Principle of analysis of two-phase cocurrent gas-liquid flow, different flow patterns and their transitions. Homogeneous and separated flow models. Void fraction. Drift flux model. Frictional pressure drop analysis, methods of Lockhart –Martinelli, Chisholm, separate-cylinders, Chenoweth-Martin correlation. Integral analysis of annular flow pattern. Vapour-liquid flow in heat transfer. Fluid-particle flow systems. Case studies on multiphase flows through evaporators and pipelines with and without heat flux, a case study on air-lift pumps.		
List of Text Books/ Reference Books/Open sources	1. Introduction to Multiphase Flow: Basic Concepts, Applications and Modelling by George Yadigaroglu, Geoffrey F. Hewitt. 2. Dynamics of Multiphase Flows by Chao Zhu, Liang-Shih Fan, Zhao Yu.		
Course Outcomes (COs):	The students of the course should be able to – CO1: Define multiphase flow and related terminologies (K1) CO2: Interpret the fundamental laws, correlations and principles of multiphase flows (K2) CO3: Analyze and apply various homogeneous and separated multiphase flow models (K4) CO4: Apply laws and principles of multiphase flow for various case studies (K3)		
CO-PO Mapping			

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1										
CO2	3	2	1									
CO3	1	3	2	1	1							
CO4	2	3	1									

Program Elective	Course Code: Che/PE/B/T/414C	APPLIED STATISTICS FOR CHEMICAL ENGINEERING (ELECTIVE-I)	Credits: 3
	Total Contact Hours: 36		
Prerequisite			
	Course Content		**Hours**
	Fundamentals of Statistical Analysis Concept of Histogram, Frequency polygon and other pictorial representations; Central Tendency; Dispersion measure; Testing of hypothesis: Z, t, F, Chi-square distributions; Annova Analysis of variant; Contingency Test & goodness of fit; Non parametric method (H, U); Nonlinear least square technique – parametric sensitivity; Problems related to Chemical Processes. Case study on usage of statistical analysis in Chemical and allied Processes.		
List of Text Books/ Reference Books/Open sources	1. Applied Statistics and Probability for Engineers by Douglas C Montgomery, George C Runger 2. Statistical Methods by N. G Das 3. Miller & Freund's Probability and Statistics for Engineers by Richard Johnson		
Course Outcomes (COs):	**The students of the course should be able to** CO1: Understand and describe key concepts and techniques used in applied statistics. (K2, K1) CO2: Discuss and assess the problems related to sampling and hypothesis testing. (K2, K3) CO3: Differentiate and critically apply parametric and non-parametric tests in Statistics. (K4, K3) CO4: Develop ability to analyse data using different statistical methods to solve problems associated in Chemical Engineering. (K5, K4)		
CO-PO Mapping			

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO 1	2	3	3												1
CO 2	2	3	2	3	2										2
CO 3	2	3	3	2	3		1					2			3
CO 4	3	3	3	2	2		1					2	1		3

Program Elective	Course Code: Che/PE/B/T/414D	Chemical Process Synthesis (ELECTIVE – I)	Credits: 3
	Total Contact Hours: 36		
Prerequisite			
	Course Content		**Hours**
	Synthesis of Steady State Flow Sheets ● Introduction ● Flow Sheets ● Basic Steps in Steady State Flow Sheeting ● Proportional Logic ● Deduction Theorem of Proportional Logic ● Resolution-Based Flow Sheet Synthesis for Chemical Plants Synthesis of Steady State Pipeline Networks ● Basic Hydraulic Equations ● Equations of Head Loss ● Equivalent Length ● Hardy Cross Method ● Linear Theory Method ● Newton-Raphson Method ● Steady Flow Analysis of Pipe Networks Synthesis of Heat Exchange Networks ● Introduction ● Concept of Pinch Technology ● Concept of ΔTmin, Composite Curve, Grand Composite Curve, Utilities, Maximum Energy Recovery ● Temperature Interval Diagram (TID) Method ● Pinch Design Method ● Synthesis of Heat Exchange Network Synthesis of Steady State Separation Systems ● Introduction ● Task of Separation ● Combination of Separation and Mixing for Minimum Separation Load ● Sequencing of Distillation Columns ● Heuristics and Designing of Distillation Systems for Multicomponent Separation Synthesis of Steady State Reactor Systems ● Types of Reaction Systems ● Choice of Reactor Model, Reactor Performance, and Reactor Conditions ● Reactor Configuration ● Selection of Reactors ● Synthesis of Reactor System ● Optimization of Complex Reactor network		
List of Text Books/ Reference Books/Open sources	1. Chemical Process Design and Integration by Robin Smith. 2. Product and Process Design Principles by Warren D. Seider 3. Chemical Process Synthesis and Engineering Design by Anil Kumar 4. Systematic Methods of Chemical Process Design by Biegler, Grossmann and Westerberg		
Course Outcomes (COs):	The students of the course should be able to: CO1: Understand the process flow sheet and synthesis of resolution based flow sheet synthesis for chemical plant CO2: Develop the **Steady State Pipe-line Networks for distribution lines** CO3: Understand the concept of Pinch Technology and develop the Heat Exchanger Networks for chemical plant CO4: Understand the task of separation in a separation system and develop flowsheet for multicomponent separation systems CO5: Develop and optimization of complex reactor network		

CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1		3	3		2		2		2					1	2
CO2		3	3		2		2		2					1	2
CO3		3	3		2		2		2					1	2
CO4		3	3		2		2		2					1	2
CO5		3	3		2		2		2					1	2

Program Elective	Course Code: Che/PE/B/T/414E	Principles of Micro-fluidics (ELECTIVE – I)	Credits: 3
	Total Contact Hours: 36		
Prerequisite			
	Course Content		**Hours**
	Overview of general microfluidics and applications: Origin, Definition, Uses, Commercial outputs, science of miniaturization. Hydrodynamics: Intermolecular forces; Continuum assumption; Governing equations; Boundary conditions; Slip theory; Scaling laws and their effects in microfluidics; Low Reynolds no. flows & examples; Liquid and particle handling in microscale. Transports in microscale (Capillary flows) & Review: Surface tension and interfacial energy; Young-Laplace equation; Contact angle; Capillary length and capillary rise; Interfacial boundary conditions, Marangoni effect. Project work on Capillary Length and capillary rise. Electrohydrodynamics: Fundamentals; Electro-osmosis, Debye layer, Thin EDL limit, Ideal electro-osmotic flow, Ideal EOF with back pressure, Cascade electroosmotic micropump, EOF of power-law fluids. Electrophoresis of particles; Electrophoretic mobility; Electrophoretic velocity dependence on particle size. Microfabrications (hard & soft lithography) : Fabrication technique overview: Current Trends. Biomedical applications with case studies in Drug delivery, Diagnostics, Bio-sensing. Microfluidics components: Micropumps: Peristaltic pumps, Rotary pumps, Centrifugal pumps, Ultrasonic pump, EHD pump, MHD pumps. Microflow sensors: Differential pressure flow sensors, Drag force flow sensors. Droplet generators: Kinetics of a droplet, Dynamics of a droplet. Microreactors: Design considerations, Liquid-phase reactors; Design consideration for PCR reactors. Project work on designing a microreactor meant for renewable production of fuels.		
List of Text Books/ Reference Books/Open sources	1. Tabeling,P., Introduction to microfluidics, Oxford University Press Inc., 2005. 2. Kirby,B.J., Micro- and Nanoscale Fluid Mechanics: Transport in Microfluidic Devices, Cambridge University Press, 2010. 3. Colin, S., Microfluidics, John Wiley & Sons, 2009.		
Course Outcomes (COs):	The students of the course should be able to – CO1: Define microfluidic system and related terminologies (K1) CO2: Interpret the fundamental laws and principles (K2) CO3: Analyze micro-scale transport of fluid under various fields using microfluidic principles (K4) CO4: Apply laws and principles of microfluidics for manufacturing simple devices (K3)		
CO-PO Mapping			

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1										
CO2	3	2	1									
CO3	1	3	2	1	1							
CO4	2	3	1									

Program Elective	Course Code: Che/PE/B/T/414F	Introduction to Nano-science & Engineering (ELECTIVE – I)	Credits: 3
	Total Contact Hours: 36		
Prerequisite			
	Course Content		**Hours**
	1. Synthesis Methodologies Pertaining to Top-Down Approach: ● Lithographic Process and Its Limitations ● Ball Milling ● Case Studies on Different Top-Down Approaches 2. Synthesis Methodologies Pertaining to Bottom-Up Approach: ● Plasma Arc Discharge ● Sputtering and Its Advantages ● Chemical Vapor Deposition and Its Types ● Molecular Beam Epitaxy ● Sol-Gel Technique: Chemistry and Examples ● Template-Based Growth of Nanomaterials ● Vapor-Liquid-Solid (VLS) Method ● Group Discussion on Various Bottom-Up Techniques and Case Studies 3. Characterization Tools of Nanomaterials: ● Concept of Atomic Force Microscopy (AFM) ● Details of Electronic Microscopy ● Spectroscopic Insights (Raman Spectroscopy, X-Ray Photoelectron Spectroscopy) 4. Introduction to Nanostructured Materials: ● Introduction to Carbon Nanotubes and Fullerenes ● Graphene: Synthesis, Applications ● Quantum Dots ● Nanocomposites ● Metal and Metal Oxide Nanowires ● Intelligence Engraved in Silicon.		
List of Text Books/ Reference Books/Open sources	1. Introduction to Nanoscience and Nanotechnology, by KK. Chttopadhyay and A.N. Banerjee 2. Principles of Nanoscience & Nano Technology, by M. A. Shah, Tokeer Ahmad 3. Introduction To Nano Science And Nano Technology, by Dr. A. Deevan Paul Mr. J. Panduranga Rao Dr. Martha Srinivas Mrs. Mandapati Lavanya 4. Nano-engineering In Science And Technology: An Introduction To The World Of Nano-design: 6 (Series on the Foundations of Natural Science and Technology, by Michael Rieth 5. Nanotechnology: An Introduction by Jereny Ramsden 6. Nanotechnology: Technology Revolution of 21st Century by Er. Rakesh Rathi		
Course Outcomes (COs):	CO 1 Define the Top Down and Bottom up approach of nanomaterial synthesis. To Describe, concept of Homogeneous Nucleation, Synthetic methodologies pertaining to Top-down approach emphasising on Lithographic process and its limitations, Ball milling (K1 and K2) CO2 Understand and develop the synthesis methodologies for nano materials like, Plasma Arc Discharge, Sputtering and its advantage, Chemical Vapour Deposition and its types, Molecular Beam Epitaxy (K2 and K3) CO3 Analyse and Compare different synthesis methodologies like, Sol-Gel Technique, Template-based growth of Nanomaterials (K4) CO4 Categorise various characterization techniques of nano materials (K5) CO5 Justify application of nanomaterials and nanostructured materials for specific applications. (K6)		

CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	**3**	**3**	**3**	**3**	**2**								**3**	**1**	
CO2	**3**	**3**	**3**	**3**	**2**								**3**	**1**	
CO3	**3**	**3**	**2**	**3**	**2**								**2**	**1**	
CO4	**2**	**1**	**1**	**3**	**2**								**2**	**2**	
CO5	**2**	**1**	**3**	**3**	**3**										

Sessional 11	**Course Code: Che/PS/B/S/411**	**CHEMICAL ENGINEERING PROJECT II**	**Credits: 2**
	Total Contact Hours:		
Prerequisite			
	Course Content		**Hours**
	Specific projects/designs will be assigned by the teachers concerned		
Course Outcomes (COs):	Students will be able to CO1apply chemical engineering knowledge to address a specific research /design problem CO2 assimilate the existing knowledge base for specific research /design problem CO3 incorporate new ideas to create originality in the specific research /design problem CO4 write report based on experimental results/ design solutions for the specific research /design problem.		

CO-PO Mapping

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PS O 2	PS O 3
CO1	3	3	3	3	3						3		3	3	3
CO2	3	3	3	3									3	3	3
CO3	3	3	2	3									1	3	3
CO4	3	3	1	2	3				3			3	1	1	3

Program Laboratory	**Course Code: Che/PC/B/S/412**	**PROCESS EQUIPMENT DESIGN AND DRAWING**	**Credits: 2**
	Total Contact Hours:		
Prerequisite			
	Course Content		**Hours**
	Pipe flow network, Pressure vessel, Filtration unit, sedimentation unit, centrifuge, cyclone separator, agitated vessel **(PEDD-I)** Distillation column, Cooling tower, Liquid-liquid extractor, Absorption column, Multi effect Evaporator, Heat exchanger network (HEN), Dryer Design, Heterogeneous Reactor **(PEDD-II)**		
Course Outcomes (COs):	CO1. Apply chemical engineering knowledge and skills for development of comprehensive design package for a chemical plant (K4, K5,K6) CO2. Screen and evaluate technology options, perform material and energy balances, develop PFD, P&ID and plantwide control strategy with considerations for safety, operability and environment compliance of the plant (K3,K4,K5) CO3. Detailed process and mechanical design of major equipment and apply computer-aided design software(s) for design and simulation of a chemical plant.(K4,K5) CO4. Develop interpersonal teamwork and communication skills to function effectively as a process engineer.(K3)		

CO-PO Mapping

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO 2	PSO 3
CO1	**3**	**3**	**2**	**2**							**1**		**3**	**2**	
CO2	**3**	**3**					**2**						**2**	**3**	
CO3	**3**	**3**	**2**		**2**								**3**	**2**	
CO4						**2**		**2**	**3**	**2**	**1**				

Sessional 12	Course Code: Che/PS/B/S/413	SEMINAR – I	Credits: 1.5
	Total Contact Hours:		
Prerequisite			
	Course Content		**Hours**
	Seminar presentation on Industrial training/Internship and research topics		
Course Outcomes (COs):	The students will be able to CO1: Demonstrate knowledge acquired during industrial training/industrial (or) research internship/project. (K2, K3, K4) CO2: Develop communication skill. (K1, K6) CO3: Recommend/Discuss solution to pertaining queries. (K2, K4, K5)		
CO-PO Mapping			

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	3		3			3		2	3	3	3
CO2										3					
CO3	3	3	2			2	3			2		3	3	3	3

Sessional 13	Course Code: Che/PC/B/S/414	PROCESS INSTRUMENTATION & CONTROL LAB	Credits: 1.5
	Total Contact Hours:		
Prerequisite			
	Course Content		**Hours**
	Testing of the characteristics of Pneumatic Control Valve		
	Measurement of liquid level by pressure bulb type level indicator and recorder		
	Testing of a Pressure Gauge		
	Studies on a Bubbler type Liquid level Measuring Device		
	Studies on the Dynamics and Two-position Control of a level Control Set up		
	Studies on the dynamic and Control of Pressure of an Air Reservoir		
	Determination of Dynamic Model from the Response Characteristics of a Thermocouple		
	Dynamics and Temperature Control of a Hot Air Blower (Process Trainer)		
Course Outcomes (COs):	CO1. Apply the knowledge and skills of chemical engineering, process instrumentation & control for analysis and tuning of various types of static and dynamic control systems. (K4, K5, K6) CO2. Apply appropriate software tools to study the static & dynamic characteristics of various control systems. (K6) CO3. Demonstrate the importance and application of appropriate instrumentation for the efficient design of process control loops for process engineering plants. (K2) CO4. Develop interpersonal teamwork and communication skills to function effectively as a process engineer.(K3)		
CO-PO Mapping			

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	1		3						3		3		
CO2	3	3			3								2		
CO3	3	3	1	2	1		2						3		
CO4	3	3				3		2	3	1	2				

Program Laboratory	**Course Code: Che/PC/B/S/415**	**Chemical Process Simulation Laboratory**	**Credits: 1.5**
	Total Contact Hours:		
Prerequisite			
	Course Content		**Hours**
	Solving Chemical Process Simulation Problems Using Different Blocks of Chemical Engineering Process Simulator (ASPEN PLUS, HYSYS, DWSIM) on: (i) Thermodynamic Property Calculation Using Equation of State (ii) Fluid Transporting Devices: Pumps, Compressor, Expander (iii) Heat Transfer Equipment: Heat Exchanger, Evaporator (iv) Mass Transfer Equipment: Flash Separator, Absorber, Distillation Column (v) Reactors		
	Case Study on Integrated Chemical Process		
Course Outcomes (COs):	CO1. Hands-on experience on handling Chemical Engineering process simulator, CO2. To generate and analyze thermodynamic and transport properties CO3. To simulate and analyze different chemical unit operations and transport processes CO4. To apply the know-how to simulate integrated chemical process		
CO-PO Mapping			

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3	3					3		
CO2	3	3	3	3	2							
CO3	3	3	3	3	2							
CO4	3	3	3	3	3					3		

CHEMICAL ENGINEERING DEPARTMENT: BACHELOR OF CHEMICAL ENGINEERING

4TH YEAR 2ND SEMESTER

Program Honors	Course Code: Che/PC/H/T/421	CHEMICAL PROJECT ENGINEERING & ECONOMICS	Credits: 4
	Total Contact Hours: 48		
Prerequisite			
	Course Content		**Hours**
	Introduction to Chemical Plant Design; Functions of Project Engineer. Process selection and evaluation; Essential Flow Diagrams for understanding chemical processes - Block Flow Diagrams (BFD), Process Flow Diagrams (PFD), Piping and Instrumentation Diagram (PID). Plant Layout - Location of Chemical plant, raw materials and utilities. Project Costs - Capital Costs and Manufacturing Costs - their estimation. Engineering Economic Analyses - Return on Investments and the Time Value of Money, Simple and Compound Interest, Discrete and Continuous compounding, Inflation, Inflation-adjusted interest rates. Cash Flow Diagrams - Discrete and Cumulative. Annuities - their present and future worth, discount factors. Depreciation of Capital Investment - Type of depreciation. Taxation, Cash Flow and Profit. Profitability Analysis of Projects and Equipments - Discounted and non-discounted profitability criteria, Incremental Economic analysis, Evaluation of Process Equipment alternatives, Incremental Analysis for Retrofitting Facilities - Discounted and non-discounted methods. Methods of cost calculations of chemical process equipment. Optimum design of pipe line, lagging thickness, heat exchanger, distillation column, reactor, storage vessels, evaporators etc. Scale-up - Introduction; Principles and theory; Concept of similarity - different types of similarity. Scale equations for common chemical engineering systems - applications and limitations. Application of Scale-up techniques to the selection and specification of chemical process equipment - solid-liquid separations, reactors, heat exchangers etc. Limitations of scale-up.		
List of Text Books/ Reference Books/Open sources	1.Plant Design and Economics for Chemical Engineers by Max S. Peters, Claus D. Timmerhaus. 2.Analysis, Synthesis and Design of Chemical processes by Richard Turton, Richard C. Bailie and Wallacce B. Whiting.		
Course Outcomes (COs):	**Students will be able to** CO1: To understand, develop, and analyze various types of flow diagrams. (K2, K3, K4 & K6) CO2: To understand and analyze economics of chemical processes with particular emphasis on various methods of cost estimation for the overall project as well as for individual equipment. (K2, K4 & K5) CO3: To understand and apply profitability and financial analysis, methods for decision making among alternatives for chemical processes including incremental analysis for retrofitting facilities. (K2, K3 & K4) CO4: To understand, apply, and integrate the basis of optimum design and scale-up of major and auxiliary chemical equipment. (K2, K3 & K5)		
CO-PO Mapping			

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3
CO1	3		3	3	3					3		2	3		3
CO2	2				3					3		3	2	3	
CO3	3				3					3		3	2	3	
CO4	2	3	3	3	3		1					3	3	3	

Program Honors	Course Code: Che/PC/H/T/422	CHEMICAL PROCESS SAFETY & RISK MANAGEMENT	Credits: 3
	Total Contact Hours: 36		
Prerequisite			
	Course Content		**Hours**
	The Importance of Process Safety; Terminology, Safety Assurance, Safety Culture, Safety Assessment, Safety in Design and Operation. The Concept of Risk – Definitions; Accepted and Imposed Risk; Perception of Risk, Acceptance Criteria, ALARP, Cost Benefit Analysis. Measurement, Calculation and Presentation of Risk Estimates for Risk Indices, Individual Risk and Societal Risk. Consequence Analysis – Source and Effect Models Source Models – Discharge rate models; Flash and Evaporation; Dispersion Models; Explosions and Fires – Vapour Cloud Explosions (VCE), Flash Fires, Physical Explosions, BLEVE and Fireball; Confined Explosions, Pool Fires, Jet Fires. Effect Models – Toxicity Effects, Thermal Effects, Explosion Effects. Event Probability and Failure Frequency Analysis –Frequency Data from historical records; The logic tree approach - Fault Tree Analysis (FTA) and Event Tree Analysis (ETA). Quantification of Logic Trees – Introduction, Fractional Dead Time, Dependent Failure Analysis, Application of Failure Data to Fault Trees; Quantification of Event Trees. Common Cause Failure (CCF) Analysis; External Events Analysis (EEA); Failure Mode and Effect Analysis (FMEA) – Methodology of FMEA, Criticality Analysis; Corrective Action and Follow-Up. HAZOP – Introduction, Basic Concepts, Conducting HAZOP studies. Human Reliability Analysis (HRA) - The role of the Operator, Control System Design, Human Error Assessment Methods, Application of HAZOP to Human Reliability; THERP – (Technique for Human Error Rate Prediction), HEP – Human Error Probability; HRAET – Human Reliability Assessment Event Tree.		
List of Text Books/ Reference Books/Open sources	1.Process Safety Analysis by an Introduction by Bob Skelton. (IChemE, UK). 2.Guidelines for Chemical Process Quantitative Risk Analysis – AIChE – CCPS (Center for Chemical Process Safety).		
Course Outcomes (COs):	**Students will be able to** CO1: Students will be able to understand hazards associated with Chemical Process Industry CO2: In case of accidents, students will know about measures to be taken CO3: Students will be able to know the hazards associated with transportation of dangerous chemicals and necessary steps to be taken for combating the hazard CO4: Students will be able to TAKE precautionary measures during design of chemical process plants.		
CO-PO Mapping			

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	2											
CO3	2											
CO4												

Engineering Science	Course Code: Che/ES/B/Mech/T/423	INDUSTRIAL MANAGEMENT	Credits: 3
	Total Contact Hours: 36		
Prerequisite			
	Course Content		Hours
	Growth of Industries, Management thoughts and scientific management, Taylorism; Factory system of production, Introduction to management problems, Types of manufacture, Planning analysis and control aspects in industries. Types of business ownership, means of finance and business combinations, organization structures, committee organization, authority and responsibility, duty and span of control. Plant location, factory buildings and physical facilities, plant layout, tools and techniques of plant layout, materials - handling arrangements. Product development, standardization, simplification and diversification. Functions of production, planning and control, production forecasting, production scheduling and network techniques, Gantt chart, CPM, PERT etc. Work study, job evaluation and merit rating; purchase system and inventory control. Inspection and quality control of systems, statistical quality control, maintenance and replacement policies for machine and equipments; decision making theories, breakeven analysis cost benefit analysis, evaluation of financial and managerial efficiencies. Introduction to operational research techniques. Application of fuzzy logic in modern management concepts. Human relations in industry and labour compensation. Personnel management, provision of industrial legislations in India. Wage and salary administrations. Welfare and safety provisions, trade union acts. Study of environmental impacts and environmental laws.		
List of Text Books/ Reference Books/Open sources			
Course Outcomes (COs):	The students of the course should be able to – CO1: Classify different components of organizations and their management roles (K2) CO2: Solve the various problems of production planning and control (K3) CO3: construct and solve basic optimization problems (K3) CO4: Analyse the fundamental concepts of decision making for financial issues (K4). CO5: Illustrate different problems related to human resources (K3)		
CO-PO Mapping			

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3		1			2		2	1		2	
CO2	2	2	1	1	1	1	1				3	
CO3	2	3	1			1				1	1	
CO4			3	2	1	1		2			1	
CO5						3	2		1			

Program Elective (H)	Course Code: Che/PE/H/T/424A	HIGH POLYMER ENGINEERING (ELECTIVE –II)	Credits: 3
	Total Contact Hours: 36		
Prerequisite			
	Course Content		**Hours**
	Fundamental concepts of High polymers, classification of High polymers, polymerization methods, kinetics Addition, Condensation and Ring opening polymerization, Reactor Design, **Case study on polystyrene reactor designing for single pass conversion with a goal of optimization of total cost** Determination of Molecular weight and Molecular weight distribution, **Designing Polymer Properties via precisely defined and tunable molecular weight distribution.** Rubbery State and compounding of natural and synthetic rubber, classification of polymer forming operations, Principles of Injection Moulding, Extrusion, Calendering, Blow moulding, Lamination. **Group Discussion on Plastic Extrusion Process and its defects.**		
List of Text Books/ Reference Books/Open sources			
Course Outcomes (COs):	**Students will be able to** CO1: (a) Describe the basic concept of polymerization and classify different types of polymers; (b) generalize the mechanism, kinetics and thermodynamics associated with various kinds of polymerization techniques. (K2) CO2: Apply the conditional knowledge to assess and solve the problems related with Emulsion, addition and condensation polymerizations. (K3) CO3: (a) Describe the Molecular weight distribution, discuss the Polymerization processes and determination of molecular weight of polymer via different techniques; (b) Interpret the glass-rubber transition and viscoelastic behavior of Polymer and solve the problems associated with the viscoelasticity by the help of viscoelastic models. (K2) CO4: (a) Illustrate the principles of Polymer processing like Extrusion, injection molding, Calendering, Blow molding etc.; (b) explain the basics of membrane fabrication techniques. (K2) CO5: Describe the techniques associated with Industrial polymer manufacturing. (K2)		

CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3										
CO2	3	3		3								
CO3	3	2		3								
CO4	3	2	3				2					
CO5	2	1	3				2					

Program Elective (H)	Course Code: Che/PE/H/T/424B	PETROLEUM REFINERY ENGINEERING AND PETROCHEMICALS (ELECTIVE-II)	Credits: 3
	Total Contact Hours: 36		
Prerequisite			
	Course Content		**Hours**
	Origin, formation and composition of petroleum. Evaluation of crude oil. Properties and testing of petroleum products. Pretreatment of crude oil and crude oil distillation. Fuel quality management of transportation fuels. Secondary conversion processes in petroleum refineries – cracking, alkylation, isomerisation, polymerization, hydrocracking, coking, visbreaking. Treatment and purification of petroleum products. Corrosion in refineries. Design of pipe-still heaters and multicomponent distillation systems. Group Discussion on Refinery-Petrochemical Complex Interface Scope of petrochemical industry, Feedstock identification – conventional and alternate.Group Discussion on feed purity and trace components, Synthesis gas and chemicals from petroleum hydrocarbons.Case Study on addition of inert feed material to control exothermic or reversible reaction, Group Discussion on production of benzene from toluene through disproportionation and hydrodealkylation, Polymers of olefins. Synthetic fibres. Synthetic rubber. Plastics, synthetic detergents, and petroleum coke & carbon black.		
List of Text Books/ Reference Books/Open sources			
Course Outcomes (COs):	**Students will be able to** CO1 Evaluate crude oil and explain Primary processing K6 and K5 CO2 Develop secondary processing and formulation K3 CO3 Determine safety measures and Routine maintenance in hydrocarbon industries K4 CO4 Appraise Scope of petrochemical industry and identify conventional and alternate Feedstock for production of Synthesis gas and chemicals K4& K6 CO5 Analyse production of Polymers of olefins including fibres, rubber, Plastics, detergents and carbon black K4		
CO-PO Mapping			

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3	3	3									
CO3	2	3	3	3								
CO4	2	3	3	3								
CO5	3	3	2	3								

Program Elective (H)	Course Code: Che/PE/H/T/424C	OPERATIONS RESEARCH (Elective II)	Credits:
	Total Contact Hours: 36		
Prerequisite			
	Course Content		**Hours**
	Overview of operations research. **(A) Deterministic Methods:** Introduction to Linear Programming - The Simplex Algorithm, Dual Simplex Method, Primal -Dual computations, sensitivity analysis. Transportation Model and its variants - Transportation algorithm, the Assignment model -the Hungarian method, the Transshipment model and case studies. Network Models and Algorithms - Scope and definitions, Minimal spanning tree algorithms, Shortest route problem, Maximal flow model. Group discussion on the Algorithms and application Advanced Linear Programming - Efficient Computational algorithms, Duality, parametric linear programming, Karmarkar Interior Point algorithm. Goal Programming - Problem Formulation, Algorithms- Weighting Method and Preemptive Method. Integer Linear Programming - illustrative applications (Sequencing - processing 'n' jobs through 'm' machines; the Travelling Salesman problem), Algorithms - Branch & Bound (B&B), Cutting-Plane, Zero-One Implicit Enumeration Algorithms. Deterministic Dynamic Programming - Forward and backward recursion, applications -cargo-loading model, work force size model, equipment replacement model, investment model, inventory models. Application of the dynamic programming in Students' project work Deterministic Inventory Models - Static and Dynamic 'Economic Order Quantity' (EOQ) models. **(B) Stochastic Methods:** Forecasting - Moving averages, exponential smoothing, regression. Decision Analysis - Decision-making under certainty, under risk and under uncertainty. Game Theory - Introduction, Optimal solution of two-person zero-sum games, solution of mixed strategy games and case studies. Probabilistic Dynamic Programming - A game of chance, the Investment Problem. Probabilistic Inventory Models - Continuous Review, Single-Period and Multi-Period models. Queueing Theory - role of exponential distribution, pure birth and death models, generalized Poisson queueing model, specialized Poisson queues. Simulation Modelling - Monte Carlo Simulation, Discrete Event Simulation. Markovian Decision Processes.		
List of Text Books/ Reference Books/Open sources	1. Operations Research by Hamdy A. Taha 2. Introduction to Operations Research by Billy E. Gillet 3. Principles of Operations Research by H. H. Wagner.		
Course Outcomes (COs):	**Students will be able to:** CO1: **Identify**, **formulate** and **analyze** models to optimize organizational resources. (K1,K4) CO2: **Define** and formulate and **solve** linear programming problems using appropriate techniques, **interpret** the results obtained and translate solutions into directives for action. (K1, K3) CO3: **Develop** mathematical skills to analyse and **solve** integer programming and network models obtained from different applications.(K4,K6) CO4: **Analyze** the performance data obtained from scheduling, queueing, and inventory and take decisions under uncertainty & risk. (K4)		

CO-PO Mapping

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO 1	2	3	3									3			
CO 2	2	3	3	3								2			2
CO 3	2	3	3	2	2							3			
CO 4	2	3	3	3		3	2				2	3		1	2

Program Elective (H)	Course Code: Che/PE/H/T/424D	BIOENERGETICS AND BIOPROCESS ENGINEERING (ELECTIVE –II)	Credits: 3
	Total Contact Hours: 36		
Prerequisite			
	Course Content		**Hours**
	Biochemical reaction engineering, Kinetics of homogeneous reactions, reaction mechanism, Temperature dependency from Arrhenius law, Interpretation of batch kinetic data, Kinetics of enzyme catalyzed reaction in free and immobilized states, Michaelis – Menten kinetics of substrate utilization, product formation and biomass production, Monod growth model and its various modifications with special reference to inhibition by substrate, product and toxic materials, structured and unstructured kinetic rate model, thermal death kinetics of cells and spores, bioreactors for suspension and immobilized cultures, solid-state fermentation, protein-protein and DNA-protein interaction, basics of genetic engineering, Chemical mechanisms of biological energy conversion in mitochondria and chloroplasts, application of bioprocesses in energy generation(generation of bioethanol, biodiesel, biogas etc., microbial fuel cell), introduction to algal biotechnology, factors affecting algal growth, photo bioreactors, photosynthesis in high density medium.		
List of Text Books/ Reference Books/Open sources	1. Bioprocess Engineering Principles, Pauline M Doran, Elsevier Science & Technology **2.** Bioprocess Engineering: Basic Concepts, Michael L. Shuler& Fikret Kargi , Pearson Publication		
Course Outcomes (COs):	**Students will be able to** CO1: Define and describe the basic mechanisms of biochemical reaction engineering, kinetics of homogeneous reactions, Michaelis -Menten kinetics, Monod growth model and its various modifications, cell growth kinetics. K1 and K2 CO2: Study the segregated and non-segregated models; reaction engineering of genetically modified cell cultures, recombinant DNA technology. K3 CO3: Describe the types of bioreactors for suspension and immobilized cultures, solid state fermentation, protein-protein and protein-DNA interaction. K4 CO4: Assessment of bioprocesses in energy generation, algal biotechnology, photosynthesis. K5& K6 CO5: Predict the efficacy of pretreatment methods of lignocellulosic biomass for bioethanol and renewable chemicals synthesis. K6		

CO-PO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3	3	3									
CO3	2	3	3	3								
CO4	2	3	3	3								
CO5	3	3	2	3								

Program Elective (H)	Course Code: Che/PE/H/T/424E	COMPUTATIONAL FLUID DYNAMICS (ELECTIVE-II)	Credits: 3
	Total Contact Hours: 36		
Prerequisite			
	Course Content		**Hours**
	OVERVIEW OF CFD: Role of CFD, Problem solving in CFD, Components of CFD Softwares GOVERNING EQUATIONS, Complete Navier-Stokes Equations, Complete Energy EquationsComplete Mass conservation Equations, Parabolized Navier-Stokes Equations, Euler Equations, Turbulence Modeling COSERVATION LAWS AND CONSERVATION FORM OF EQUATIONS Conservative and non conservative forms, Source terms and boundary conditions, Approximation of Governing Equations, Finite difference methods, Finite Volume methods Finite Element methods GRID GENERATION Structured Grid generation, Unstructured Grid generation, Adaptive Grid generation SOLUTION ALGORITHMS AND TECHNIQUES LU decomposition, Approximate factorization, Relaxation algorithms, Hybrid Schemes CFD METHODS FOR THE EULAR EQUATION Linearization and Jacobian Matrix, Eigenvalues and Eigenvectors, Flux splitting methods CFD METHODS FOR NAVIER-STOKES EQUATIONS Beam-Warming algorithm, MacCormack's scheme, Upwind Techniques PRESENTATION OF CFD RESULTS INTRODUCTION PARALLEL COMPUTING Case study on solution for an incompressible Couette flow using an implicit method and the pressure correction method		
List of Text Books/ Reference Books/Open sources	1. Computational Fluid Dynamics by J. D. Anderson, McGraw-Hill. 2. Computational Fluid Mechanics and Heat Transfer by J.C. Tannehill, D.A.Anderson, R.H. Pletcher, Taylor and Francis. 3. Computational Fluid Dynamics by T. J. Chung, Cambridge University Press.		
Course Outcomes (COs):	1. To understand the basic structure and function of CFD code 2. To formulate CFD problems applying different techniques viz Finite difference, Finite volume and Finite element 3. Solve conservation equations (Euler equation, Navier Stokes Equation) using different CFD techniques 4. Apply CFD knowledge to address Chemical Engineering problems		
CO-PO Mapping			

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	2	3	3	3								
CO3	3	3	3	3								
CO4	3	3	3	3						2		

Program Elective (H)	Course Code: Che/PE/H/T/424F	ENVIRONMENTAL BIOTECHNOLOGY (ELECTIVE II)	Credits: 3
	Total Contact Hours: 36		
Prerequisite			
	Course Content		**Hours**
	Basics of microbiology related to Environmental Biotechnology, Biofilm Kinetics, Suspended Floc, Dispersed Growth and Slurry Reactors, Fixed Film Reactors, Attached Growth Reactors, Immobilized Cell Reactors, Activated Sludge Process, Bulking and other Sludge Settling Problems, Lagoons, Aerobic Biofilm Processes, Trickling Filter and Biological Towers, Rotating Biological Contactors, Granular Media Filters, Fluidized Bed Circulating Bed Biofilm Reactors, Hybrid suspended growth/Biofilm processes, Nitrification. ANAMMOX process, Denitrification, Anaerobic Treatment by methanogenesis, Detoxification ofhazardous chemicals, Bioremediation, Ex-situ and In-situ applications., Degradation of polymers, Degradation of dyes, Tannery effluent, Semiconductor waste treatment, treatment of pulp and paper industry effluent, Desulfurization and aromatic removal of petroleum fractions. Bioremediation of water, soil and air.		
List of Text Books/ Reference Books/Open sources	1. Environmental Biotechnology theory and Application, Evans G M, John Wiley 2. Environmental Biotechnology, Bruce E Rittmann And Perry L Mccarty, Mc Graw Hill India 3. Environmental Biotechnology, Alan H Scragg, Longman		
Course Outcomes (COs):	Students will be able to CO1: To understand principles of microbiology related to environmental biotechnology. (K2) CO2: To understand biotechnological application for the environmental issues (K2, K3) CO3: To understand various remediation processes and its application to control pollution (K2, K3) CO4: Able to Understand the design concepts and selection of bioreactors for pollution control (K3,K4)		
CO-PO Mapping			

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	1				3	3						3		
CO2	3	2											3	3	
CO3	3	3					3						3		
CO4	3	2	3				3						3	1	

Program CORE	Course Code: Che/PE/H/T/424G	MODERN ENERGY ENGINEERING AND ENERGY MANAGEMENT (ELECTIVE-II)	Credits:
	Total Contact Hours: 36		
Prerequisite			
	Course Content		**Hours**
	Concept of "waste to energy"; Thermo-chemical and biochemical conversion of Biomass; Kinetics of incineration, Pyrolysis and Gasification (low and medium Joule) of Biomass and polymeric Wastes; Design of incinerator, Green House effect of Incineration; Design of Updraft, downdraft and fluidized bed Gasifiers for Biomass. Design of low temperature pyrolyser and Fast/Flash Pyrolyser; bio-diesel, Kinetics of Biogas generation from wastes; Design of Bio-digesters for the generation of Biogas; **Case study on biogas generation from cow dung.** Gas Turbines and Dual-fuel IC engines for power generation from low and medium Joule gases; use of Natural gas for the substitution of liquid fuels; Dehydration and Desulfurization processes of natural gas; gathering and transport of natural gas and Biogas; concept and operation of fuel cell. Energy Management & Energy Planning. Definition & Objectives of Energy Management. Duties and responsibilities of energy managers and auditors. Energy Audit and Energy Monitoring. Integrated energy management─ case studies. Maximizing System Efficiency		
List of Text Books/ Reference Books/Open sources			
Course Outcomes (COs):	The students will be able to CO1: Recognize potential of waste to produce energy through various processes CO2: Understand and design various bioenergy generation processes and equipment CO3: Recognize and analyse importance of energy conservation and management		
CO-PO Mapping			

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3		3				3						3		
CO2	3	2	3	2			3						3	3	
CO3	3	3	3		2		3						3		3

Sessional 13	Course Code: Che/PC/B/S/421	CHEMICAL PLANT DESIGN AND DRAWING	Credits: 1.5
	Total Contact Hours:		
Prerequisite			
	Course Content		**Hours**
	Problem based on integrated chemical processes		
List of Text Books/ Reference Books/Open sources			
Course Outcomes (COs):	**Course Outcome** CO1. Apply chemical engineering knowledge and skills for development of comprehensive design package for a chemical plant (K4, K5,K6) CO2. Screen and evaluate technology options, perform material and energy balances, develop PFD, P&ID and plantwide control strategy with considerations for safety, operability and environment compliance of the plant (K3,K4,K5) CO3. Detailed process and mechanical design of major equipment and apply computer-aided design software(s) for design and simulation of a chemical plant.(K4,K5) CO4. Develop interpersonal teamwork and communication skills to function effectively as a process engineer.(K3)		
CO-PO Mapping			

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO 2	PSO 3
CO1	3	3	2	2							1		3	2	
CO2	3	3					2						2	3	
CO3	3	3	2		2								3	2	
CO4						2		2	3	2	1				

Program Laboratory	**Course Code: Che/PC/B/S/422**	**HEAT & MASS TRANSFER LABORATORY**	**Credits:**
	Total Contact Hours:		
Prerequisite			
	Course Content		**Hours**
	Determination of heat transfer coefficient		
	Unsteady state heating in a jacketed vessel		
	Studies on heat loss from a pipe		
	Studies on heat exchanger performance		
	Evaporator performance		
	Refrigerator performance		
	Determination of mass transfer coefficient		
	Experiment on batch distillation		
	Studies on distillation column performance		
	Humidification column performance		
	Dehumidifier performance		
	Drier performance		
	Determination of drying rate		
List of Text Books/ Reference Books/Open sources			
Course Outcomes (COs):	**Course Outcome** CO1 interpret the experimental results based on theoretical knowledge of heat and mass transfer CO2 visualize the equipment and its operation studied in theoretical classes CO3 demonstrate hands on experience of the operation of the equipment		
CO-PO Mapping			

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	3	3	2	2	2			2	2	2		2	3	2	
CO2					2				2			2	2	3	
CO3								2	3	3		2	3	2	

Program Laboratory	Course Code: Che/PS/B/S/423	SEMINAR – II	Credits:
	Total Contact Hours:		
Prerequisite			
	Course Content		**Hours**
	Seminar presentation based on research works		
Course Outcomes (COs):	The students will be able CO1: Acquire knowledge during project work on various types of state of the art research projects (or) Home designs. (K2, K4, K5, K6) CO2: Develop communication skill. (K1, K6) CO3: Recommend/Discuss solution to pertaining queries. (K2, K4, K5)		
CO-PO Mapping			

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	3		3						3	3	3
CO2										3					
CO3	3	3	2			2	3			2		3	3	3	3

Program Laboratory	Course Code: Che/PC/B/S/424	GENERAL VIVA-VOCE	Credits:
	Total Contact Hours:		
Prerequisite			
	Course Content		**Hours**
	All chemical engineering subjects		
Course Outcomes (COs):	Student will be able to CO-1: assess of theoretical knowledge in chemical engineering domain (K3) CO-2: design and develop chemical engineering process and equipment (K5) CO-3: appraise and assimilate fundamental chemical engineering knowledge for chemical industry (K6)		
CO-PO Mapping			

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	2	2					2			3	2	
CO2	3	3	3	2	2					2			2	3	
CO3	3	3	3	3	2					2			3	2	

Open Elective	**Course Code: FET/OE/CHE/T/101**	**CATALYTIC PROCESSES FOR GREEN ENERGY**	**Credits: 3**
	Total Contact Hours: 36		
Prerequisite			
	Course Content		**Hours**
1.	Concept of Green Energy; role of catalysis for sustainable green energy. Catalysts: Definition, types, mode of action, activity, selectivity, stability, classification, comparison between homogeneous and heterogeneous catalysts.		3
2.	Preparation of solid catalysts: Bulk and supported metal, metal oxides/complexes catalysts, nano- catalysts, undoped and doped semiconductor based photocatalysts; promoters.		4
3.	Characterization of catalysts: Physicochemical characterization – Chemical composition, morphology, surface area, surface acidity and basicity, BET-BJH, TGA, FTIR, XPS, XRD, HRTEM, FESEM, TPD, ICP- AES, UV-vis DRS spectroscopy, PL techniques, band gap measurements with some case studies.		5
4.	Reaction kinetics in heterogenous catalysis: -Mass and heat transfer effects, -Mechanism of surface kinetics: Langmuir and Eley–Rideal mechanisms. Biocatalysts and kinetics of enzyme-catalysed reactions.		4
5.	Deactivation, Regeneration, Recovery and Reuse of Catalysts: –Mechanisms and kinetics of catalyst decay; aging, coking and poisoning –Catalyst regeneration, recovery and reuse.		4
6.	Industrial Catalytic Processes for production of green energy: -Hydrotreatment for green petroleum fuels; - Catalysis in thermochemical conversion processes of biomass to fuel (Pyrolysis, Gasification, Biomass to liquid (BTL)) - Catalysis in conversion of biomass with reference to pretreatment and hydrolysis for fermentable sugar; Lignin conversion processes for biofuel - Catalysis in coal to liquid process (CTL) -Biodiesel and Bio-aviation fuel production. -Fuel Additive production for enhanced engine performance and harmful emission reduction -Conversion of CO2 to green fuel.		9
7.	Photocatalysts: –Design, principle and applications in water splitting, solar photocatalysis; CO2 photo reduction etc. Electrocatalysis in fuel cells: –Basic principles; types and catalyst; anodic and cathodic reaction; methanol oxidation. Other emerging processes for green energy production.		7
List of Text Books/ Reference Books/Open sources	1. Heterogeneous Catalysis - Fundamentals and Applications Julian R.H. Ross - Elsevier 2. Principles of heterogeneous catalysis – J M Thomas and W J Thomas - VCH 3. Catalysis for Green Energy and Technology- Bagheri, Samira--Springer 4. Elements of Chemical Reaction Engineering- H. Scott Fogler -PHI 5. Photocatalysis: Complete Set--Editors: Jenny Schneider, Detlef Bahnemann, Jinhua Ye, Gianluca Li Puma, Dionysios D Dionysiou-RSC Publishing 6. Catalysis for Renewables: From Feedstock to Energy Production Edited by Gabriele Centi and Rutger A. van Santen WILEY-VCH Verlag GmbH & Co.		

Course Outcomes (COs):	**Course Outcome** CO1 To **define** green energy and **describe** the basic aspects of catalysis for sustainable application (K1) CO2 To **understand** the synthesis of specific catalysts and **evaluate** the related physico-chemical properties (K2+K5) CO3 To **determine** reaction kinetics of heterogeneous and biocatalytic reactions (K4) CO4 To **outline** deactivation, regeneration, recovery and reuse of catalysts (K4) CO5 To **explain** the application of industrial catalytic processes for the production of green energy and **appraise** their performance (K6) CO6 To **design, develop** and **assess** photocatalysis, electrocatalysis and other emerging processes (K5+K6)
CO-PO Mapping	

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3	3	3									
CO3	2	3	3	3								
CO4	2	3	3	3								
CO5	3	3	2	3								
CO6	3	2	3	3								

Open Elective	Course Code: FET/OE/CHE/T/201	Colloids and Surface Engineering	Credits: 3
	Total Contact Hours: 36		
Prerequisite			
	Course Content		**Hours**
Module 1:Colloid &Interface, Curvature and Capillarity	Introduction to colloids: their origin, classification and application, colloidal dispersion, stability of a colloidal system.		**4**
	Surface and Interface molecular origin, the work of cohesion and adhesion. Theory of capillarity – Young Laplace equation- shape of an interface		**3**
	Chemical equilibrium across a curved interface, Generalized Kelvin equation, Oswald ripening Capillary condensation, Curvature and nucleation		**3**
	Surfactants, Concept of critical micelle concentration, thermodynamics of micellization		**2**
Module 2:Dynamics of the surfaces	Interaction forces and potential, chemical and physical interaction, classification of physical forces. Van der Waals force, electrokinetic forces		**4**
	Wetting of solid surfaces Young's equation, Young Dupre equation, contact angle and its measurement, dynamic contact angle, wetting and adhesion, approximation of surface tension of a solid		**4**
	Dynamics of liquid interface , capillary penetration, spreading, Concept of Thin films and its application.		**2**
	Wetting by solutions containing surfactants, marangoni flow.		**3**
Module 3: Engineering Applications	Catalysis and adsorption, Energetics of adsorption; Surface modification;		**2**
	Emulsions, micro emulsions and foams; Impregnation, floatation		**2**
	Detergency; enhanced petroleum recovery; microelectronics, Coatings, Biological interfaces		**2**
	Nano materials: Preparation, lithographic techniques, applications.		**3**
List of Text Books/ Reference Books/Open sources	1. Authors (Year) – Name of the text Book, Name of the Publisher 1. Principles of Colloid and Surface Chemistry, Paul C. Hiemenz, Raj Rajagopalan, Third Edition revised and expanded, 1997, CRC press, Taylor & Francis 2. Foundation of Colloid Science, second edition, Robert J. Hunter, 2000, Oxford 3. Surface, Interface and Colloids Principles and Applications, Drew Myers, 1999, John Wiley & Sons (Wiley-VCH) 4. Colloid and Interface Science, Pallab Ghosh, 2009, PHI Learning Pvt. Ltd., New Delhi 11001. 5. Physical Chemistry of Surfaces, Arthur W. Adamson, Alice P. Gast 1997, John Wiley & Sons 6. Interfacial Phenomena and Colloid Stability: Industrial Applications, Tharwat F. Tadros, Walter de Gruyter GmbH & Co KG, 01-Jul-2015 - Science - 417 pages 7. An Introduction to Interfaces and Colloids: The Bridge to Nanoscience, John C Berg, World Scientific Publishing Company, 18-Nov-2009 - Science - 804 pages 8. Colloids and Interfaces in Life Sciences and Biotechnology, 2011, 2nd Edition, Willem Norde, CRC press Taylor Francis Group 9. Interfacial Phenomena in Biological Systems, Edited by Max Bender, 1991 Marcel Dekker Inc. 10. Intermolecular and Surface Forces, Jacob Israelachvili, 3rd Edition, Academic Press, Elsevier. 11. NPTEL web or video Course NPTEL: Prof. Asutosh Sharma, Weblinkhttp://nptel.ac.in/courses/103104045/pdf_version 12. https://nptel.ac.in/courses/103/103/103103033/		

Course Outcomes (COs):	**Course Outcomes** **Students will be able to** CO1 To **understand** origin of colloids & relevance of interfacial phenomena and **appraise** the theory of capillarity and surface curvature (kinetics of surfaces) (K1+K6) CO2 To **explain** molecular origin of the intermolecular and elctrokinetic forces and **evaluate** dynamics of surfaces (K5) CO3 To **apply** basic knowledge of colloid and interfaces towards diversified fields of engineering and nanotechnology (K3).
CO-PO Mapping	

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3									
CO2	3	3	3									
CO3	2	3	3	3								